B2SPICE

EDA TOOL AND SOFTWARE

EXPERIMENTAL MANUAL

…. A MESSAGE FROM

AKADEMIKA

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For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

INDEX

FEATURES

- Redesigned user interface - toolbar, tab, and button driven
- for more intuitive control - fewer obscure menu commands
- to control program
- Virtual instruments -oscilloscope, ammeter, voltmeter, wattmeter, distortion meters,.
- “Live Circuit" parameters can be altered while simulation is running.
- Results displayed on virtual instruments or on the schematic via circuit animation
- New parts browser with customizable parts palette
- Curve tracer to plot the behavior of a device before you place
- it in your circuit
- Simulation mode - continuous transient simulation
- Multiple level tests -Multiple parameter sweeps or montecarlo sweeps in any test
- “Real-world" devices model performance of actual parts such as capacitors, resistors and inductors
- Circuit wizard to step you through the creation of the most commonly used circuit configurations
- Graphing module - complete control over all aspects of the graph Cross probing that interactively updates the graph as you add or move the probes around the circuit
- Circuit visualization to display the actual current flow through the circuit and the relative voltage relationships by varying the wire's display color
- Quick, easy, and intuitive schematic entry
- 12powerful tests
- 25,000 digital and analog parts including hundreds of realistic behavioral models for such parts as resistors, inductors and capacitors Parameterized sub circuits
- Create a part from any circuit
  Password protected defects
- Shared models
- Database editor to import and manage the library of parts
- Integrated symbol editor

INTRODUCTION

Congratulations on purchasing the best value in integrated circuit design, simulation, and analysis software. B^2Spice A/D v5 contains a mixed mode simulator based partly on the **Berkeley SPICE simulator** and partly on the **Georgia Tech XSPICE simulator**. This means that you are getting industrial strength accuracy. Built upon the proven performance of B^2Spice A/D 2000 and B^2Spice A/D v4, V5's almost complete rewrite now brings you improvements and additions that will greatly expand the program's functionality and usefulness. From new graphing to virtual instruments and scenario editors, V5 is now more powerful and cost effect than ever before.

The parts database is made up of parts from Analog Devices, Burr Brown, Com linear, Motorola, National Semiconductor, Texas Instruments, Apex, Amp, Elantec, Maxim, Linear Technology, Zetex, and many more. V5 also adds new interactive devices, such as switches and fuses that you can control and have change with the changing circuit conditions.

After you have completed the installation, we suggest you jump right into the tutorials to get acquainted with B^2Spice A/D v5.

The best way to learn a program's capabilities is to get down and dirty with it. We have a few Tutorials and we have sample circuits to get you started. Load up some sample circuits and choose Run Simulation… from the Simulation menu. Follow along with the tutorials. This will get you oriented.

Then dive into the Toolbox and Virtual Instruments. Try the parts selection from the Parts Chooser, setting up tests and virtual instruments.

Right click on components in the schematic and double click on them. These will pull up dialogs and menus. For the graph, click and double-click and control-click on plot names in the menu.

We have put a lot of work into making the program easy to use and flexible. If there's something you can't figure out how to do, drop us an email and we'll try to help.
B^2Spice A/D v5 is intended to help you design analog, digital, and mixed mode circuits. Rather than working on your circuit design with physical components, which require expensive test equipment and a lab, B^2Spice A/D v5 allows you to perform realistic simulations on your circuit without clipping wires or splashing solder. With B^2Spice A/D v5, editing and simulating circuits is a quick, easy, even enjoyable process.

B^2Spice A/D v5 supports the full Spice 3F5 set of commands, options, and models. This includes simulations such as DC Sweep, AC Sweep, Transient, Sensitivity, Pole-Zero, Fourier, Distortion analysis, and more. Models include no less than six distinct MOSFET models including BSIM3 and BSIM4, models for switches, several transmission line models, and much more.

B^{2}Spice A/D v5 is an application with two separate subprograms: the main program, and the Database Editor. The main program is most frequently used. You'll use it to create and edit your circuits, to set up the simulations, to run the simulations, and to view the results. The Database Editor is used for defining new parts or modifying those already in the parts bin. Each subprogram is covered in its own section.

The program features a large database of devices that should be sufficient for most circuits, and can be customized to meet your design needs. Some of these are catalogued in the Technical Reference section, along with the standard parameters and those available for customizing. The Database Editor section will explain how you can add more devices into the database.

B^{2}Spice A/D v5 comes in various versions. Different versions will have features and abilities that others do not need. These differences will be discussed in the user manual.

EXPERIMENT NO.1

EXPERIMENT NO: 1

NAME

Common Emitter BJT Amplifier

OBJECTIVE

The purpose of this experiment is to investigate the operation of a common-emitter BJT transistor amplifier using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number Of Components Required
BJT	2n2222	Transistor	1
RESISTOR	R	Resistor	6
CAPACITOR	C	Capacitor	3
VGEN	Vsin	Ac Voltage Source	1
VDC	Vcc	Dc Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

The Common Emitter Amplifier is one of the three basic BJT transistor amplifier configurations. In comparison to the FET common-source amplifier, the BJT amplifier has much lower input impedance, but a higher voltage gain. In this experiment, the student will build and investigate a simple, common emitter BJT amplifier. It is assumed that the student has had some background in basic transistor amplifier theory, including the use of simple ac equivalent circuits. The student is expected to develop his or her own procedure for performing the lab experiment, after having done a complete PRE-LAB analysis, and to then analyze, and thoughtfully summarize, the results of the experiment in a lab report. Additionally, the use of B^2Spice V5 as computer simulation tool is described, to further enhance the learning process.

For the circuit shown, predict the following DC parameters: IC, VCE, and VB using the "biasline technique". Recall that the operating point of the BJT can be found graphically from the intersection of the load line with the graph of output characteristics (which relates IC to VCE.

Draw an AC equivalent circuit, and predict the voltage gain (Av), input impedance Zi, and output impedance Zo, assuming all capacitors act as "ac shorts". Also draw the "composite" waveforms expected at test points Vs and Vo in the circuit, assuming the input signal Vs is a 100 mV peak, 3 kHz sine-wave.

$$A_v = \frac{A_I R_L}{Z_i} = \frac{-h_{fe} R_L}{h_{ie} + (1 + h_{fe}) R_e} \qquad \text{..............................1}$$

$$\text{Where, } A_I = \frac{-I_c}{I_b} = \frac{-h_{fe} I_b}{I_b} = -h_{fe} \qquad \text{..............................2}$$

$$A_v \approx \frac{-h_{fe}}{1 + h_{fe}} \frac{R_L}{R_e} \approx \frac{-R_L}{R_e}$$

The voltage gain calculated using the Eqn. 1 won't accurately match with the practical results because every signal source has a particular source impedance. Taking the source impedance Rs into account the overall voltage gain Avs is given by the equation below;

$$A_{vs} = \frac{V_o}{V_i} = \frac{V_o}{V_i}\frac{V_i}{V_o} = A_v \frac{Z'_i}{R_s + Z'_i} \qquad \text{..............................3}$$

where, $Z'_i = Z_i \,||\, R_1 \,||\, R_2$

Also, draw a basic "black box" model (or equivalent circuit) for a voltage amplifier, consisting of input impedance Zi, a dependent voltage source, and output impedance Zo. (Recall that Zi appears across the input of the amplifier, and Zo appears in series with the output of the amplifier's dependent voltage source).

$$Z_i = \frac{V_i}{I_b} = h_{ie} + (1 + h_{fe}) R_e \qquad \text{..............................4}$$

$$Z_o \approx \frac{1 + h_{fe}}{h_{oe}} = \frac{1}{h_{ob}} \qquad \text{..............................5}$$

In B^{2}Spice you can measure Zi by inserting a test resistor (e.g.: 1 Mega ohm) in series with the signal input to the amplifier, and measuring how much of the ac generator signal actually appears at the input of the amplifier (note the voltage divider between source and Zi in your diagram). For example, if R1 || R2 = Zi , the amplifier's input signal will be half of the applied input signal. (N.B.: the input impedance of the test instrument may affect these readings!)

You can determine Zo as follows: temporarily remove the load resistor, and measure the unloaded ac output voltage. Then replace the load, and re-measure the ac output voltage. Use these measurements to determine Zo (note the voltage divider between Zo and RL in your equivalent circuit to help see this).

LAB MEASUREMENTS/IN-LAB CIRCUIT MEASUREMENTS

Build the circuit shown, and verify ALL of the predictions above. First check the DC quiescent voltages with no ac input signal. If these are reasonably close to your predictions, connect the ac input signal to the input of the amplifier.

- Use DC coupling and dual-trace on the oscilloscope, as appropriate, to observe the waveforms at various points.
- Is the output signal (Vo) close to expected?
- How do the waveforms seen at the BJT's collector and the output (Vo) compare?
- Then temporarily remove CE, and re-measure the output voltage; did it decrease as expected? Note that if you increase the input signal, Vo will increase, along with the non-linear distortion. You can reduce this non-linear distortion by adding a small "swamping resistor" of say 100 ohms at the source.

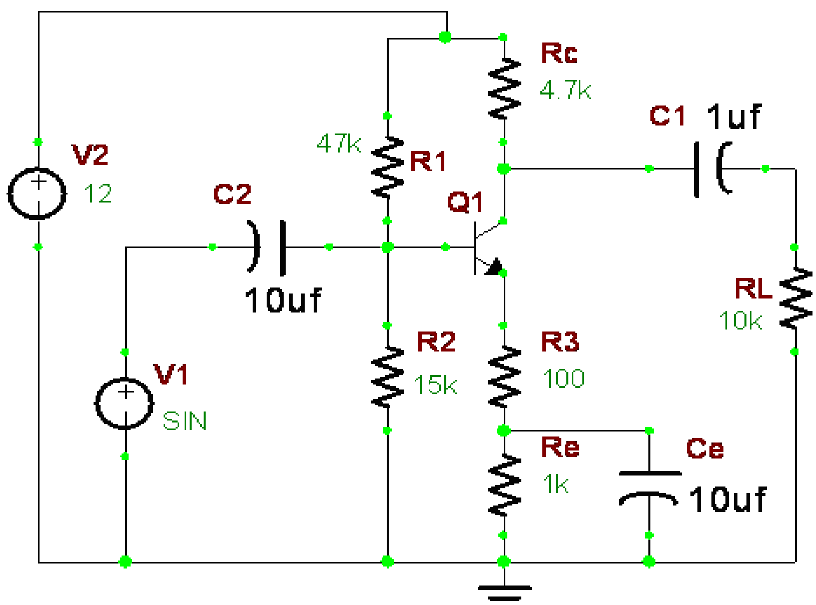


Figure 1.1: Common Emitter BJT Amplifier

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 1.1 using the editor within B^{2}Spice V5.

Step 1: Open B^{2}Spice V5 project window by clicking on the B^{2}Spice V5 project navigator shortcut on the desktop or from the path

Start Menu → All Programs → B^{2}Spice A_D V5 → B^{2}Spice V5.exe.

Figure 1.2: Open B^{2}Spice V5

Step 2: Select the required components for the design from Add Part menu just at the left side of the B2 Spice project window. The Add Part menu includes all the components available with B2 Spice A/D V5. For an easy approach of finding the required components all the elements are distributed into different types such as: Active, Digital, Passive, and Virtual components. Now find out all the elements required for your circuit and place them on the circuit schematic window.

Step 3: Select the required device from the database and place them on the circuit project file. Also at the time of selecting components one has the option of previewing the component being selected. Place all the components in their appropriate places and connect them through Circuit Lines.

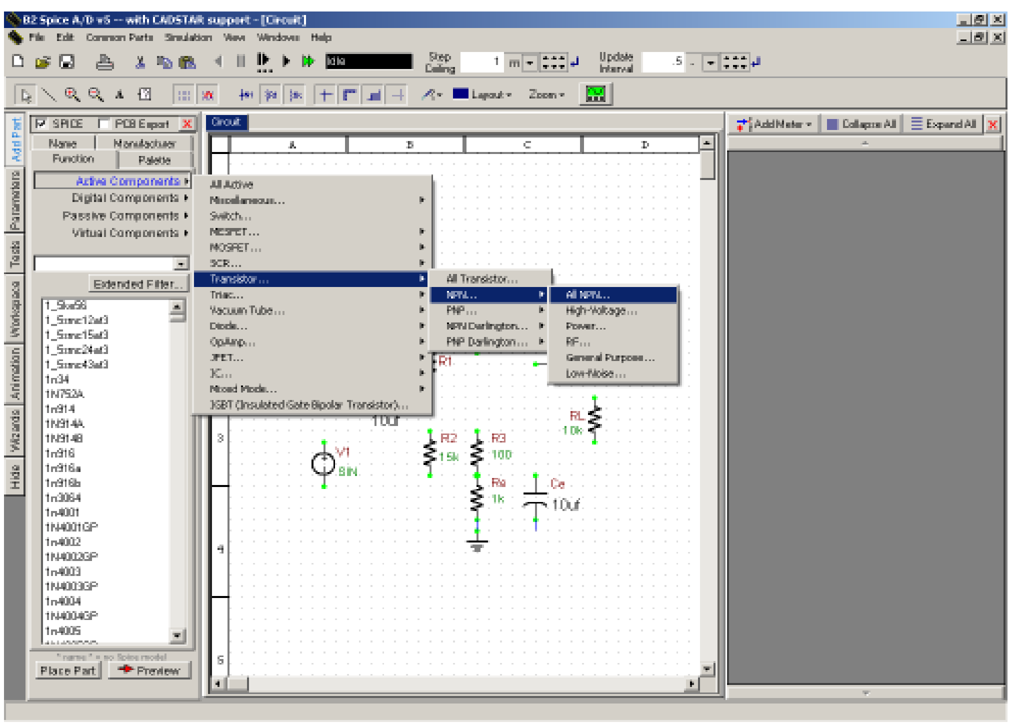


Figure 1.3: Add Components

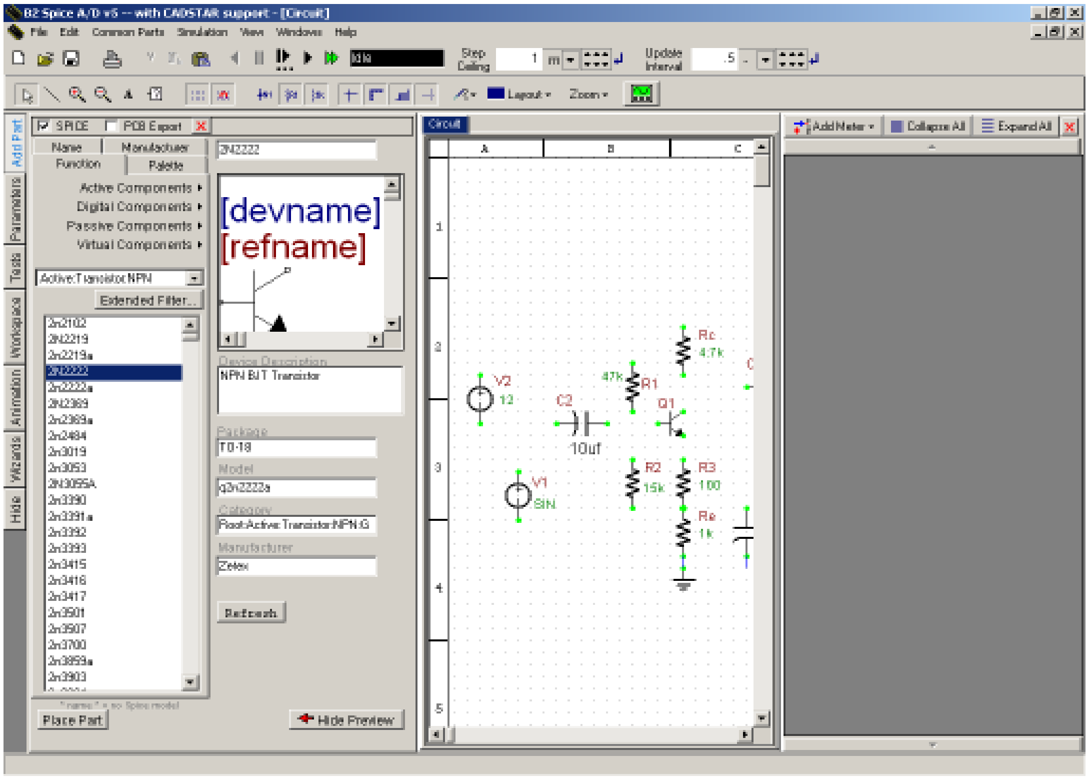


Figure 1.4: Draw the Circuit

Step 4: Press 'W' key on your key board in order to enable the Drawing of Circuit Lines. Then route with whole circuit as shown in Figure 1.1.

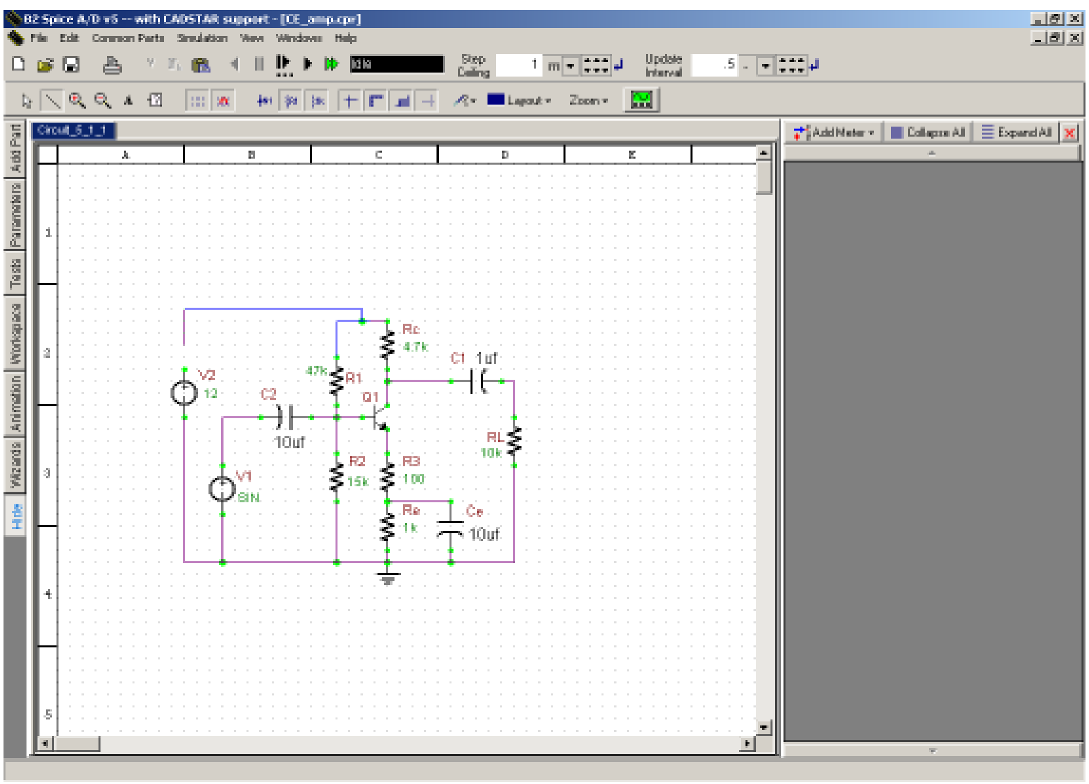


Figure 1.5: Complete Circuit

Step 5: Set the values for parameters of every device. For AC and DC voltage sources the simulation model's properties are as shown in figures below respectively.

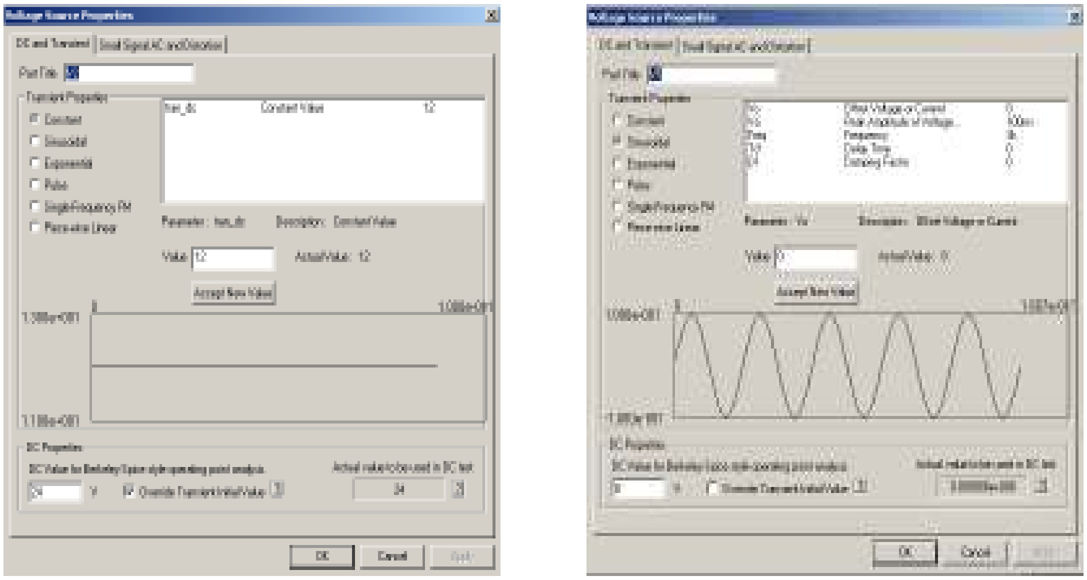


Figure 1.6: a. DC Voltage Source Properties b. AC Voltage Source Properties

For Capacitors and resistors the parameters are illustrated in below figures and for transistor the values can be seen from the simulation model property window.

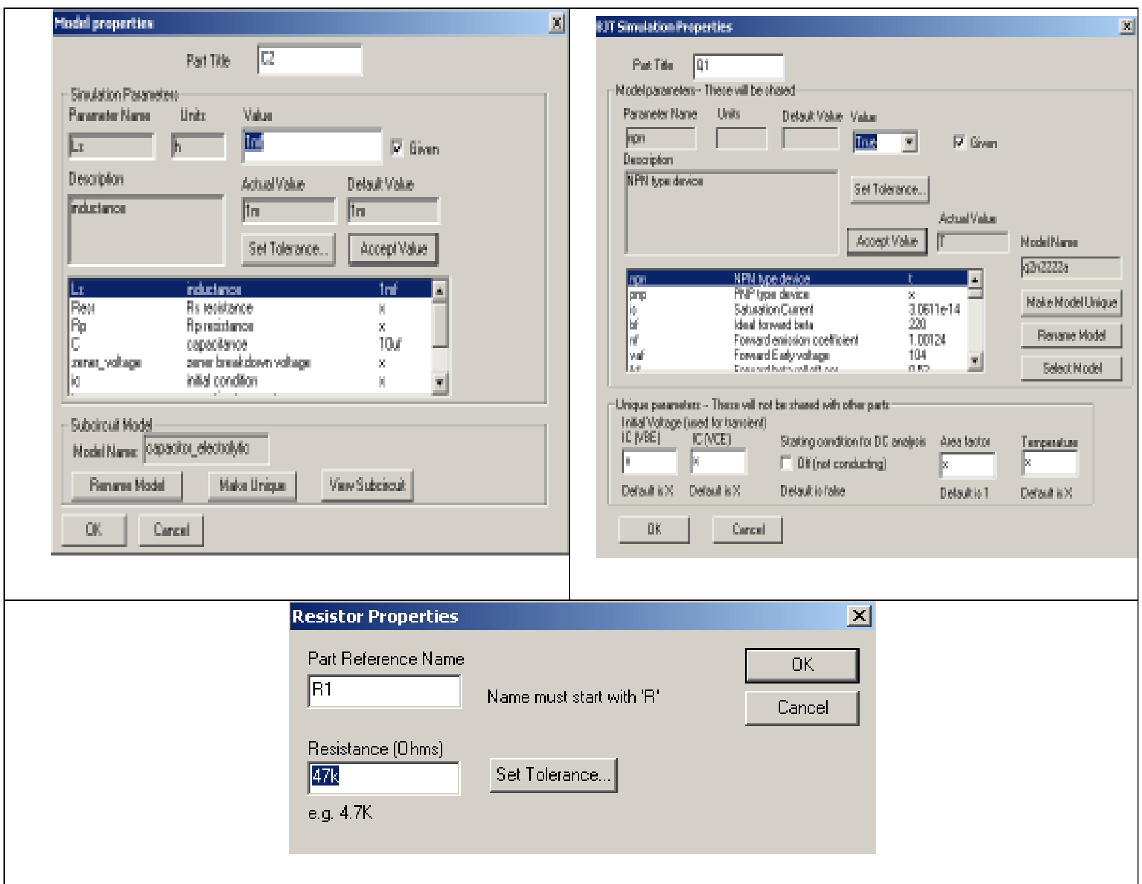


Figure 1.7: Model Properties for Resistor, Capacitor and BJT

Step 6: After completion of the circuit design go for simulation.
For simulation go to file menu → view → show instruments

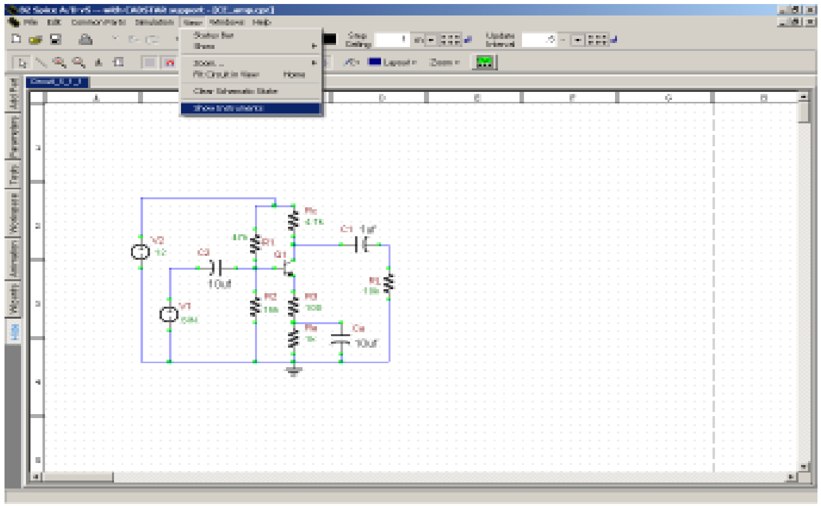


Figure 1.8: Show Instruments

Step 7: There go to Add Meters → Oscilloscope. Then you will see the Oscilloscope window.

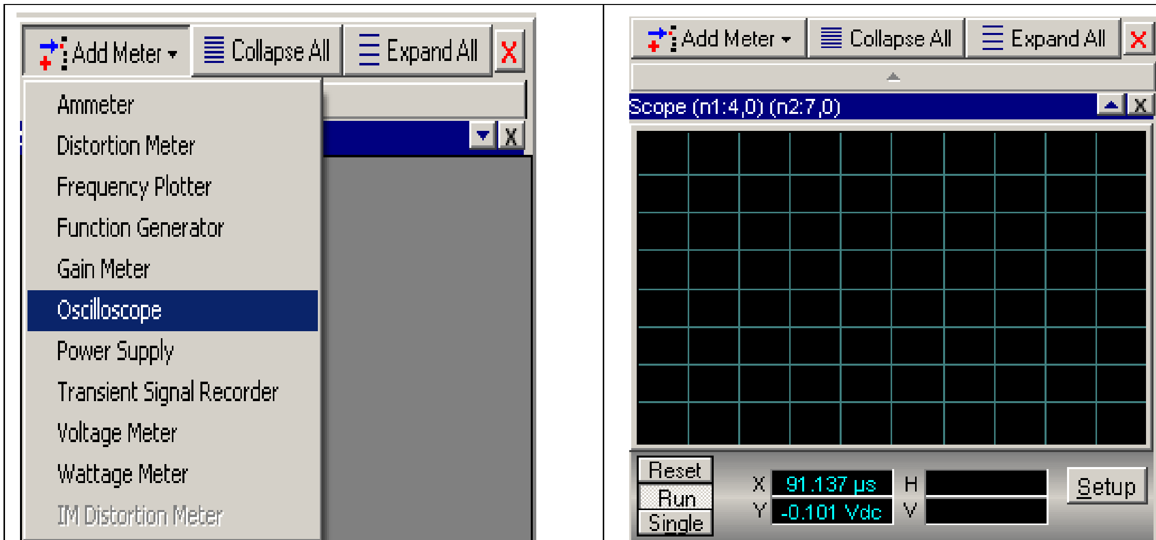


Figure 1.9: Add Oscilloscope

Step 8: Click on the Setup button on the Scope and modify the Input and Output Node numbers on Input 1 Trace and Input 2 Trace respectively with respect to the Reference Nodes. To verify and measure the gain, adjust the Gain parameter with the help of the spin buttons available to scale both the traces in the same window.

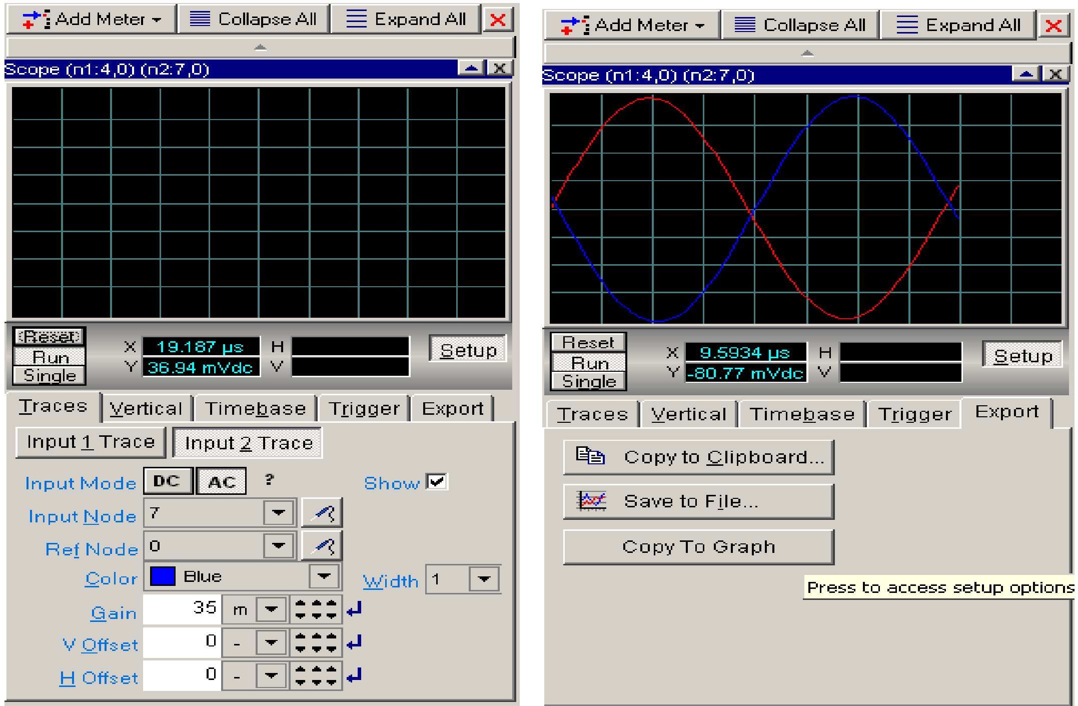


Figure 1.10: Setting up the properties for Oscilloscope

Step 9: Now to run the simulation press “F5” on keyboard or go to file menu → simulation → Click Run. Say yes for the message being displayed to adjust the step ceiling. Then to view the enlarged simulation result click on Export tab on the Oscilloscope and click on Copy to Graph as shown in above figure.

Connect the amplifier’s input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit’s component values, and see how this affects the output signal; for example, by temporarily removing Ce, you can see that the output signal dramatically decreases. By changing the emitter and collector resistors, it may be possible to further optimize the amplifier for a larger peak output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

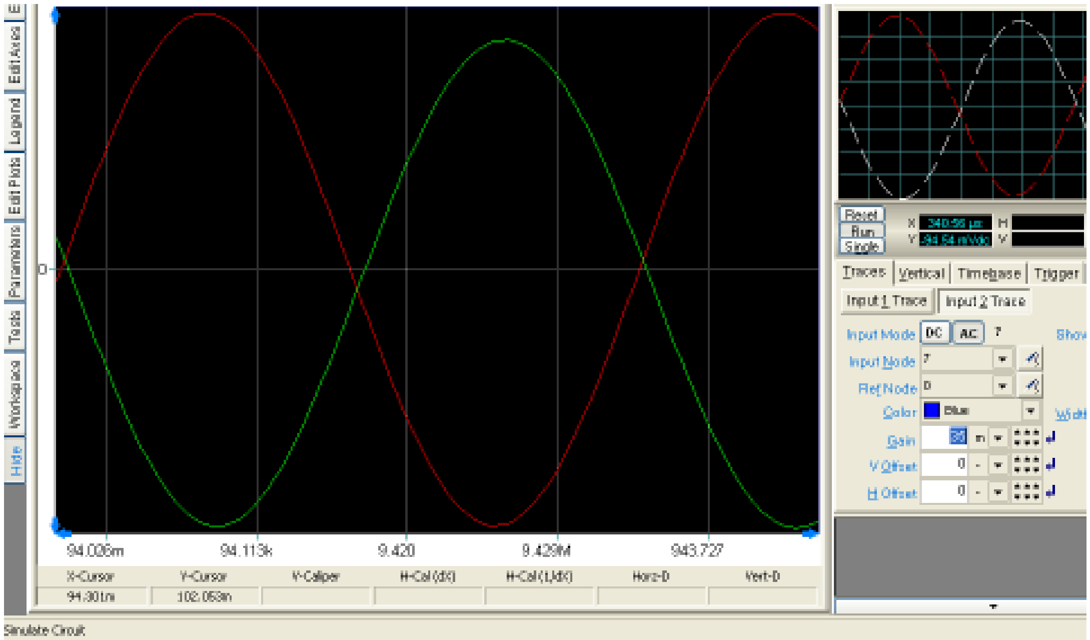


Figure 1.11: Simulation result for Common Emitter BJT Amplifier

COMMENTS AND CONCLUSIONS

Compare your results with your PRE-LAB calculations. Any significant discrepancies can be explained. Also, the purpose or function of the various components in the circuit can clearly be explained; for example, what is the purpose of the emitter bypass capacitor? Why does a swamping resistor reduce the non-linear distortion, and at what cost? How does this circuit generally compare to a common-source amplifier?

EXPERIMENT NO.2

EXPERIMENT NO: 2

NAME

Common Source JFET Amplifier

OBJECTIVE

The purpose of this experiment is to investigate the operation of a common-source JFET transistor amplifier and to perform the AC, DC, Transient and Tolerance or Sensitivity analyses of this circuit using B2Spice. Compare the same with actual performance.

CIRCUIT NOTE

Name	**B^2Spice Components Used**	**Description**	**Number of components required**
JFET	bf510	Transistor	1
RESISTOR	R	Resistor	5
CAPACITOR	C	Capacitor	3
VGEN	VSIN	AC voltage source	1
VDC	VCC	Dc voltage source	1
GND	GND	Ground	1

DESIGN AND THEORY

The Common Source Amplifier is one of the three basic FET transistor amplifier configurations. In comparison to the BJT common-emitter amplifier, the FET amplifier has much higher input impedance, but a lower voltage gain. In this experiment, the student will build and investigate a simple n-channel, common source JFET amplifier. It is assumed that the student has had some background in basic transistor amplifier theory, including the use of simple ac equivalent circuits. The student is expected to develop his or her own procedure for performing the lab experiment, after having done a complete PRE-LAB analysis, and to then analyze, and thoughtfully summarize, the results of the experiment in a lab report. Additionally, the use of B^2Spice V5 as computer simulation tool is described, to further enhance the learning process.

For the circuit shown, predict the following DC parameters: ID, VD, VG, VD, and VDS, using the "biasline technique". Recall that the operating point of the JFET can be found graphically from the intersection of the JFET "transconductance" or "transfer" equation (which relates ID to VGS) with the circuit biasline equation (VGS = - ID* RS).

Draw an ac equivalent circuit, and predict the voltage gain (Av), input impedance Zi, and output impedance Zo, assuming all capacitors act as "ac shorts". Also draw the "composite" waveforms expected at test points Vs and Vo in the circuit, assuming the input signal Vin is a 100 mv peak, 3 kHz sine wave.

The voltage gain is given by

$$Av = \frac{-gm + Ygd}{YL + Yds + gd + Ygd} \quad \text{………………………………………1}$$

Where,

- YL = 1 / ZL = admittance corresponding to ZL
- Yds = jωCds = admittance corresponding to Cds
- gd = 1 / rd = conductance corresponding to rd
- Ygd = jωCgd = admittance corresponding to Cgd
- At low frequencies the FET capacitances can be neglected.
  Under these conditions, Yds = Ygd = 0, and Eq. 1 reduces to

$$Av = \frac{-gm}{YL + gd}$$

Also, draw a basic "black box" model (or equivalent circuit) for a voltage amplifier, consisting of input impedance Zi, a dependent voltage source, and output impedance Zo. (Recall that Zi appears across the input o f the amplifier, and Zo appears in series with the output of the amplifier's dependent voltage source).

$$Zi = 1 / Yi \quad \text{………………………………………2}$$

Where, Yi = Ygs + (1 - Av) Ygd = input admittance

$$Zo = \frac{rdRd}{rd + Rd} \quad \text{………………………………………3}$$

You can measure Zi by inserting a test resistor (e.g.: 1 Mega ohm) in series with the signal input to the amplifier, and measuring how much of the ac generator signal actually appears at the input of the amplifier (note the voltage divider between source and Zi in your diagram). For example, if R1 || R2 = Zi, the amplifier's input signal will be half of the applied input signal. (N.B.: the input impedance of the test instrument may affect these readings!)

You can determine Zo as follows: temporarily remove the load resistor, and measure the unloaded ac output voltage. Then replace the load, and re-measure the ac output voltage. Use these measurements to determine Zo (note the voltage divider between Zo and RL in your equivalent circuit to help see this.

LAB MEASUREMENTS/IN-LAB CIRCUIT MEASUREMENTS

Build the circuit shown, and verify ALL of the predictions above. First check the DC quiescent voltages with no ac input signal. If these are reasonably close to your predictions, connect the ac input signal to the input of the amplifier. Use DC coupling and dual-trace on the oscilloscope, as appropriate, to observe the waveforms at various points. Is the output signal (Vo) close to expected? How do the waveforms seen at the FET's drain and the output (Vo) compare? Then temporarily remove CS, and re-measure the output voltage; did it decrease as expected? Note that if you increase the input signal, Vo will increase, along with the non-linear distortion created by the variations in the FET trans-conductance, gm. You can reduce this non-linear distortion by adding a small "swamping resistor" of say 100 ohms at the source (explain how this works). What "price" is paid for this benefit?

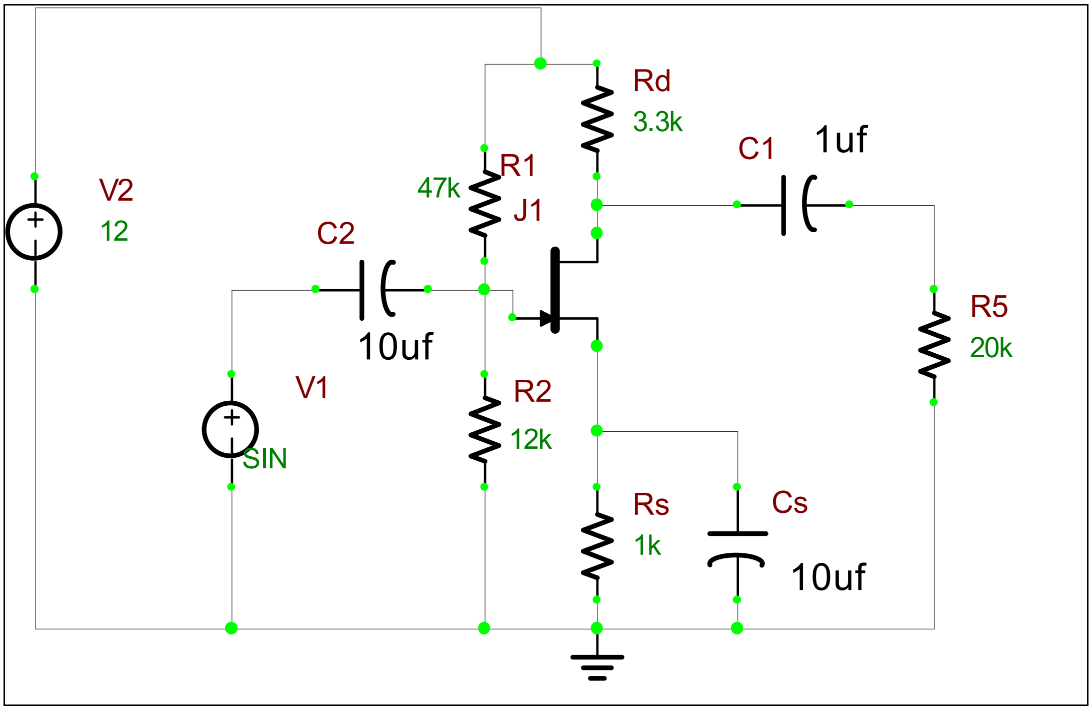


Figure 2.1: Common Source FET Amplifier

<u>COMPUTER SIMULATION USING B2SPICE V5</u>

Draw the circuit shown in Figure 2.1 using the editor within B^2Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^2Spice V5. Then connect the oscilloscope in B^2Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal; for example, by temporarily removing CS, you can see that the output signal dramatically decreases. By changing the source and drain resistors, it may be possible to further optimize the amplifier for a larger peak output signal. In summary, by using B^2Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

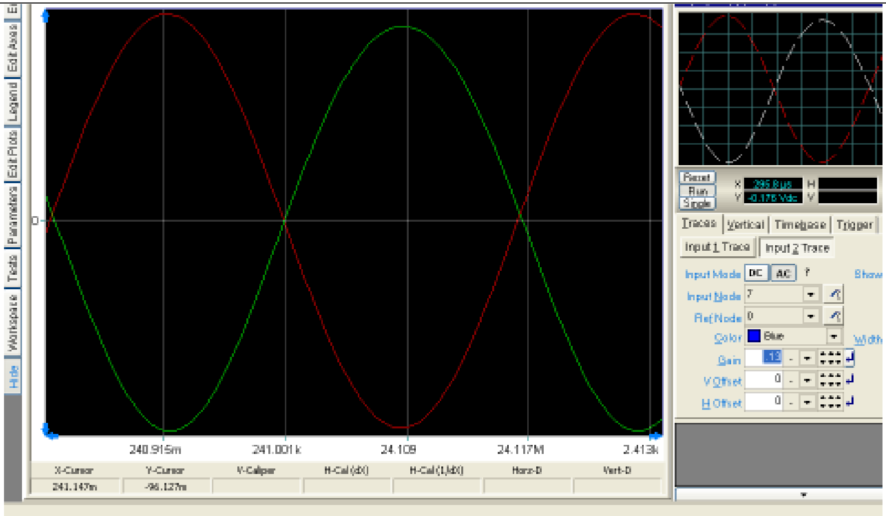


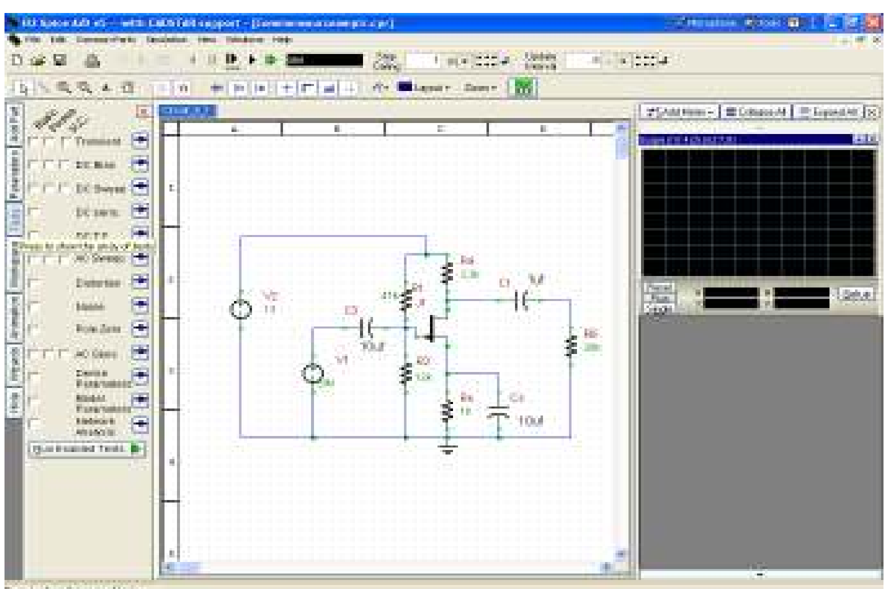
Figure 2.2: Simulation result for Common Source JFET Amplifier

For more accurate simulations in our case, we changed the ideal default JFET model in B^2Spice V5 to use the following JFET parameters, to more closely simulate the specified JFET with an IDSS = 3mA and a VGS c/o = 0.8V of bf510 JFET. For a JFET, a reasonably simple approximation for the drain current is that ID = Beta * (Vgs-Vto)2, where Beta is the FET "trans-conductance coefficient", and Vto is the FET "threshold or cutoff voltage". We changed these parameters to: "Vto" = - 0.88V, and "Beta" = 2.018mA/V2.

Here for this experiment in order to the perform the AC, DC, Transient and Tolerance or Sensitivity analyses we need to carry out few more steps after Step 9.

Step 10: Click on the Tests Tab on the left side of the project window in order to show the array of tests. Then you can perform the AC, DC, Transient or Sensitivity analyses as per required. The different analyses are parameterized and you need to provide with all the values needed. Once all parameters are set you can run the analyses.

Figure 2.3: All Tests Options



Step 11: For Transient Analyses click on the Selection Box under Basic and then press the arrow button by the side of Transient in order to specify the Test options and Transient setup.

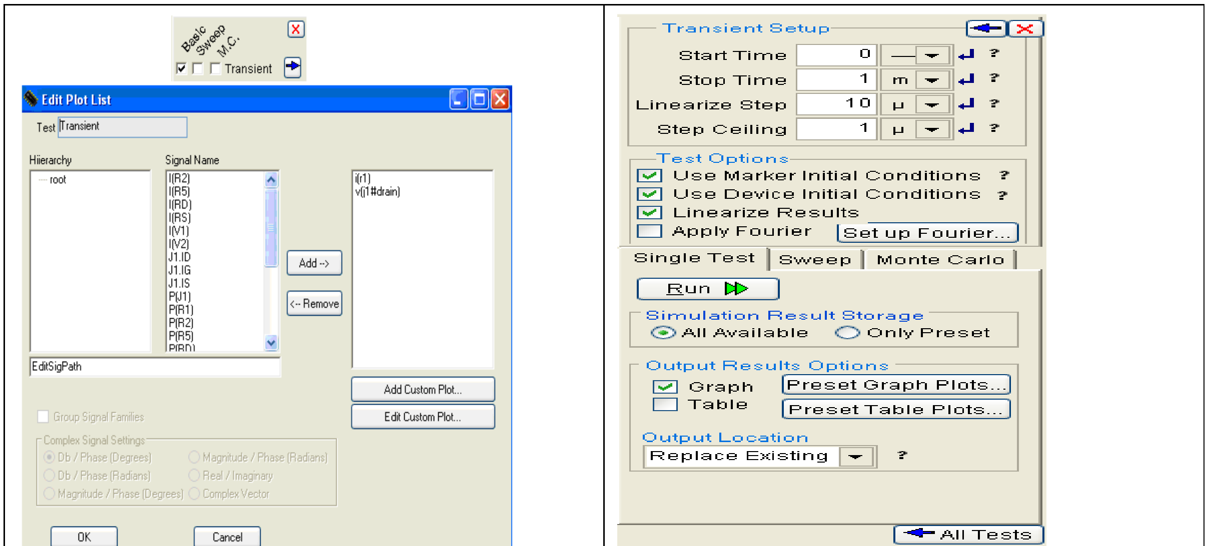


Figure 2.4: Transient Analyses Setup

Now specify the parameters for Transient Setup and select the required Test Options. Then click on Preset Graph Plots to mention the signals on which the graph to be plotted. Now click on Run to find out the Graph for Transient Analyses. To return back to all tests click on All Tests.

Step 12: Similarly you can simulate for AC, DC and Sensitivity analyses through the test arrays by providing the proper parameters in the Setup.

COMMENTS AND CONCLUSIONS

Compare your results with your PRE-LAB calculations. Any significant discrepancies can be explained. Also, the purpose or function of the various components in the circuit can clearly be explained; for example, what is the purpose of the emitter bypass capacitor? Why does a swamping resistor reduce the non-linear distortion, and at what cost? How does this circuit generally compare to a common-emitter amplifier?

EXPERIMENT NO.3

EXPERIMENT NO: 3

NAME

Casecade Or Two Stage RC Coupled Amplifier

OBJECTIVE

To design and simulate a Cascade Amplifier or Two stage RC Coupled Amplifier circuit using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number of components required
BJT	q2n3904	Transistor	2
RESISTOR	RES	Resistor	11
CAPACITOR	CAP	Capacitor	5
VGEN	VSIN	AC voltage source	1
VDC	VCC	Dc voltage source	1
GND	GND	Ground	1

DESIGN AND THEORY

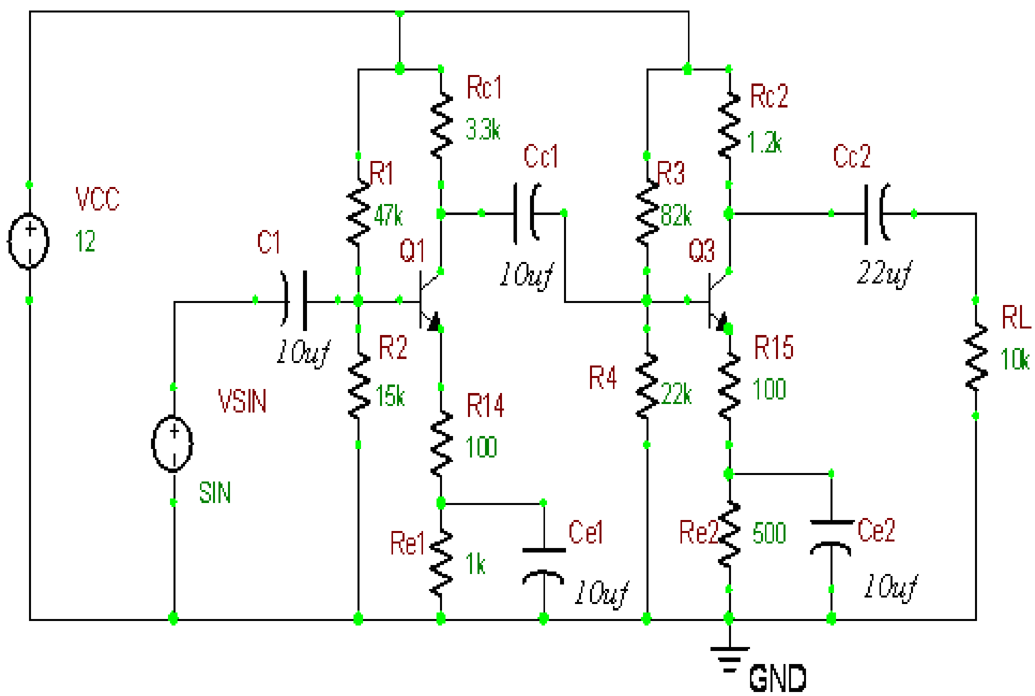


Figure 3.1: Cascade or Two Stage RC Coupled Amplifier Circuit

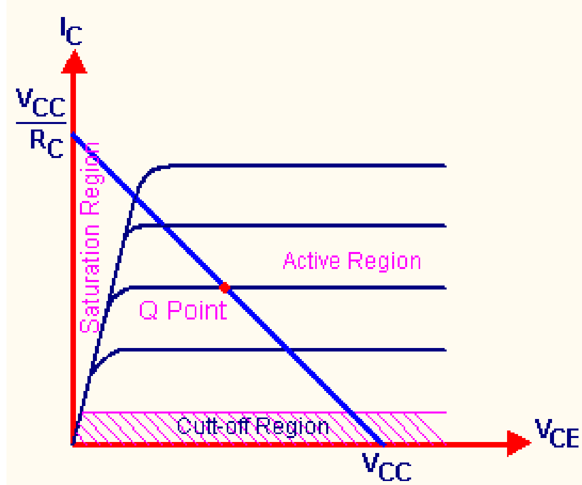
DC ANALYSIS

A transistor amplifier is designed to work in the active region. A single stage amplifier is designed using voltage divider bias. In a voltage divider bias the base voltage, VB is given by $V_B = V_{CC}\dfrac{R2}{R1+R2}$

First we bias the circuit in the active region with dc conditions.

i.e. Current through R1, IR1 is a large current of the order of 20 or 30 times that of IB.

Figure 3.2: DC Load Line to findout Q Point



Current through R2, IR2 is given by

We select V_{CC} as 12V.

Let ICQ = 2mA – ref. Manufacturer's Datasheet

$V_E = 0.2V_{CC} = 2.4\,V$

Applying Kirchhoff's Voltage Law for the output side of amplifier

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E$$

$$I_E \cong I_C - (7)$$

$$I_C R_C = V_{CC} - V_{CE} - I_E R_E$$

$$I_C R_C = V_{CC} - V_{CE} - V_E$$

$$\therefore R_C = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{12 - 5 - 1.2}{2mA} = 2.9\,k$$

We select collector resistance Rc = 3.3 k

$$\boxed{V_E = I_E R_E \cong I_C R_E}$$

$$R_E = \frac{V_E}{I_C} = \frac{0.2V_{CC}}{2mA} = \frac{2.4}{2mA} = 1.2k\Omega$$

DESIGN OF R1 & R2

For a fluctuation in I_{R1} and I_{R2} there will be small change in I_B. If I_{R1}=21I_B and a 5% change in I_{R1} occurs there will be only $5/21\%$ change in I_B. Therefore the circuit will be stable against small changes in R_1 and R_2 due to temperature or tolerance.

Thus $R_2 \leq \frac{\beta \cdot R_E}{10} - (8)$

And $V_B = V_{CC} \cdot \frac{R_2}{R_1 + R_2} - (9)$

$V_{BE} = V_B - V_E - (10)$

$\therefore V_B = V_{BE} + V_E - (11)$

=0.7+0.2VCC

=0.7+2.4 = 3.1V

From calculations, we get R2 = 15k

$V_B(R_1 + R_2) = V_{CC}.R_2$

$R_1 + R_2 = V_{CC} \cdot \frac{R_2}{V_B}$

$R_1 = V_{CC} \cdot \frac{R_2}{V_B} - R_2 - (12)$

= 47k

Similarly the second stage is designed to get the values as shown in the circuit above. The RC coupled amplifier circuit gives a phase shift of 360^0.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 3.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

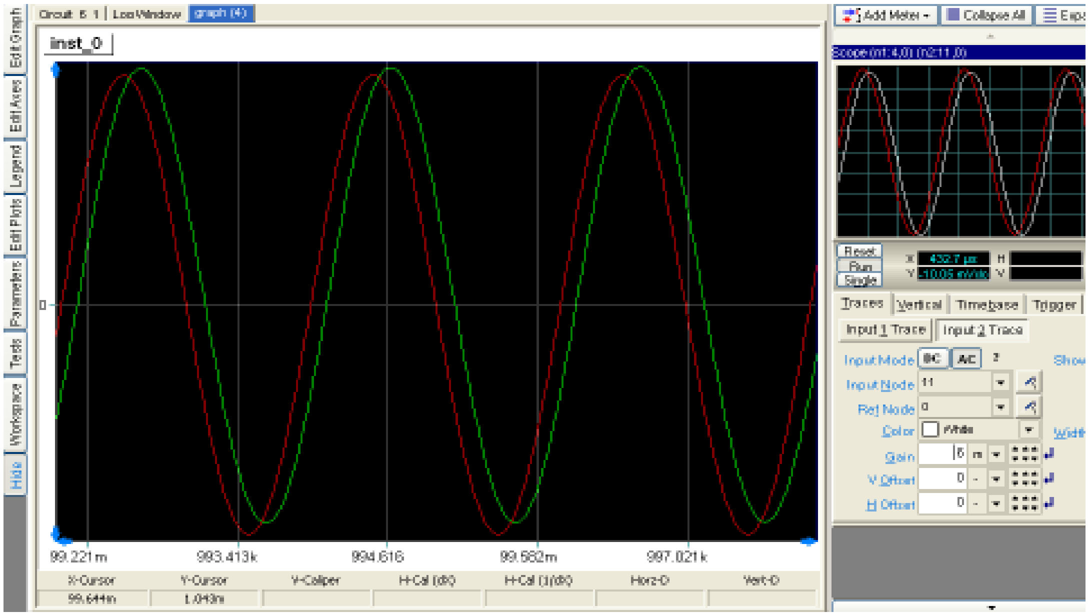


Figure 3.3: Simulation Result for Casecade or Two Stage RC Coupled Amplifier Circuit

COMMENTS AND CONCLUSIONS

Compare your results with your PRE-LAB calculations. Any significant discrepancies can be explained. Also, the purpose or function of the various components in the circuit can clearly be explained; for example, what is the purpose of the emitter bypass capacitor? Why does a swamping resistor reduce the non-linear distortion, and at what cost? How does this circuit generally compare to a single stage RC coupled amplifier?

EXPERIMENT NO.4

EXPERIMENT NO:4

NAME

Current Shunt Feedback Amplifiers

OBJECTIVE

To design and simulate a Current Shunt Feedback Amplifier circuit using B^2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number of components required
BJT	qbc547a	Transistor	2
RESISTOR	RES	Resistor	7
VGEN	VSIN	AC voltage source	1
VDC	VCC	Dc voltage source	1
GND	GND	Ground	1

DESIGN AND THEORY

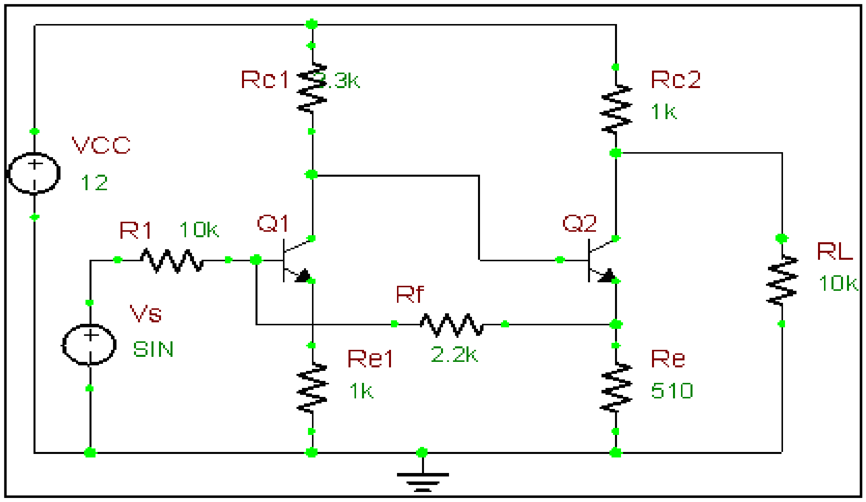


Figure 4.1: Current Shunt Feedback Amplifier Circuit

Figure 4.1 shows two transistors in cascade with feedback from the second emitter to the first base through the resistor Rf. We now verify that this connection produces negative feedback. The voltage V_{i2} is 180^0 out of phase with V_{i1}. Because of emitter follower action, V_{e2} is only slightly smaller than Vi2 and these voltages are in phase. Hence Ve2 is larger in magnitude than V_{i1} and in 180^0 out of phase. If the input signal increases so that Ig increases, If also increases, and Ii = Ig - If is smaller than it would be if there were no feedback. This action is characteristic of negative feedback.

We now show that if $R_f >> R_e$ the configuration of Figure 1 approximates a current-shunt feedback pair. Since $V_{e2} >> V_{i1}$ then

$$I_f = \frac{V_{i1} - V_{e2}}{Rf} \approx \frac{V_{e2}}{Rf} \qquad \text{.......... 1}$$

If we neglect the base current of Q2 compared with the collector current and if Rf >> R_e, then $V_{e2} \approx -I_0R_e$, and from Eq. 1,

$$I_f \approx \frac{I_0R_e}{Rf} = \beta I_0 \qquad \text{.......... 2}$$

Where β=Re/Rf. Since the feedback current is proportional to the output current, this circuit is an example of a current-shunt feedback amplifier. From the Eq. 1 and 2, and assuming Ig ≈ If,

$$A_{if} = \frac{I_0}{I_g} \approx \frac{1}{\beta} = \frac{Rf}{R_e} \qquad \text{.......... 3}$$

And hence we have verified that Aif is stable provided that Rf and Re are stable resistances.

COMPUTER SIMULATION USING B²SPICE V5

Draw the circuit shown in Figure 4.1 using the editor within B²Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

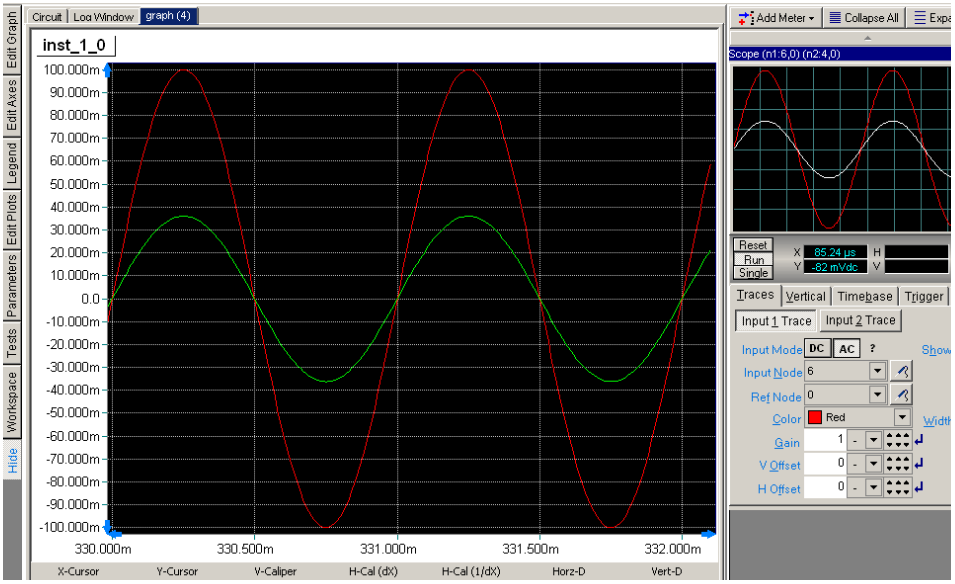


Figure 4.2: Simulation Result for Current Shunt Feedback Amplifier Circuit

Connect the amplifier's input to the AC voltage source in B^2Spice V5. Then connect the oscilloscope in B^2Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal; for example, by temporarily removing CS, you can see that the output signal dramatically decreases. By changing the source and drain resistors, it may be possible to further optimize the amplifier for a larger peak output signal. In summary, by using B^2Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

COMMENTS AND CONCLUSIONS

The above simulation result illustrates the operation of a current-shunt feedback amplifier. For a current-shunt network the voltage gain decreases. Here we expect the input resistance to be low and output resistance to be high. If we assume that Rif = 0, then Vs = Is Rs and the voltage gain with feedback is

$$A_{vgf} = \frac{V_0}{V_s} = \frac{I_0 R_{c2}}{I_s R_s} \approx \frac{R_f R_{c2}}{R_e R_s}$$

Note that if Re, Rf, Rc2, and Rs are stable elements, then Avsf is stable (independent of the transistor parameters, the temperature, or supply-voltage variations).

EXPERIMENT NO.5

EXPERIMENT NO:5

NAME

Wien Bridge Oscillator

OBJECTIVE

To design and simulate a Wien Bridge Oscillator circuit using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number of components required
BJT	q2n4401	Transistor	2
RESISTOR	RES	Resistor	8
CAPACITOR	CAP	Capacitor	4
VDC	VCC	Dc voltage source	1
GND	GND	Ground	1

DESIGN AND THEORY

Of all the low-frequency oscillator configurations, the Wien Bridge has to be the friendliest and easiest to get along with. It uses standard components, gives a good sine wave and is fairly immune to the type of op amp it is designed around.

It can, however, be misunderstood, and over simplifications as to its operation can leave the designer thinking that it is not as well trained as originally thought.

CIRCUIT

The circuit of a standard Wien Bridge oscillator is shown in Figure 5.1. A circuit will oscillate if, at a given frequency, it has greater than unity gain and zero phase shift from input, through the device, through the feedback network and back to the input.

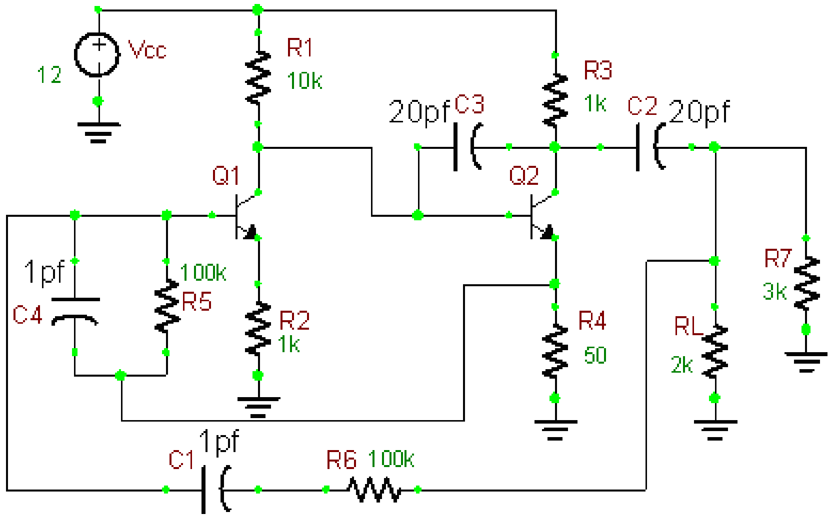


Figure 5.1: Wein Bridge Oscillator Circuit

Consider Phase Shift

Looking at the circuit in Figure 1, R6 and C1 produce a positive phase shifted current with respect to the output voltage. When this current meets R3 and C4, these components produce a voltage that is phase shifted in a negative direction. At one frequency the phase shift caused by R6 and C1 will be offset by an equal and opposite phase shift caused by R3 and C4 and the net phase shift will be zero. The circuit is now in danger of oscillating.

For the mathematically inclined, the transfer function of the network made up by R6, C1, R3, and C4 can be considered. It is relatively straightforward to derive the transfer function of the Wien Network (the resistive divider made up of R6 and C1 on the top and R3 and C4 on the bottom). The transfer function is

$$TF = \frac{jwC1R2}{1 + jw(C1R2 + C2R2 + C1R1) - w^2(C1C2R1R2)}$$

Put simply (if this is possible) the imaginary 'j' terms represent a 90° phase shift in the transfer function (either positive or negative). The real, non 'j' terms represent zero phase shift in the transfer function. As the magnitude-of the real and imaginary terms change, so does the resultant phase shift. It can be seen from above equation that at:

$$w^2 = \frac{1}{(C1C2R1R2)}$$

the real terms in the denominator equate to zero, leaving only imaginary terms in the numerator and denominator. Dividing the numerator and denominator by 'jw' leaves no imaginary terms and hence no phase shift. We therefore have our condition for zero phase shift at a given frequency.

CONSIDER THE GAIN

Once the battle between the real and imaginary terms has ceased and the resonant frequency determined, the transfer function is:

$$TF = \frac{C1R2}{C1R2 + C2R2 + C1R1}$$

Now, to greatly simplify things, it is wise to equate R6 to R3 and C1 to C4. It can then be seen that at resonance the transfer function from output to input is:

$$TF = \frac{1}{3}$$

To meet the requirements for oscillation (zero phase shift and unity gain), the circuit needs to have a gain of three to overcome the attenuation resulting from the Wien Bridge network. Looking at this another way, to keep the two inputs of the op amp at the same voltage, the resistor network from the output to the inverting input needs to provide an attenuation of three to match the attenuation of the Wien Network.

This is very convenient in theory, but near useless in practice. We can obtain resistors with accurate values but obtaining accurate capacitors is a bit more difficult. Getting capacitors with an accuracy greater than 20% starts to eat into the design budget, so it is now wise to consider the effect of different capacitor values on the performance of the circuit.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 5.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

The simulation result here is a continuous damping process where we can show them in 2 or 3 consecutive snaps of the simulation process. The below are the result of an oscillation due to the Wien Bridge Oscillator.

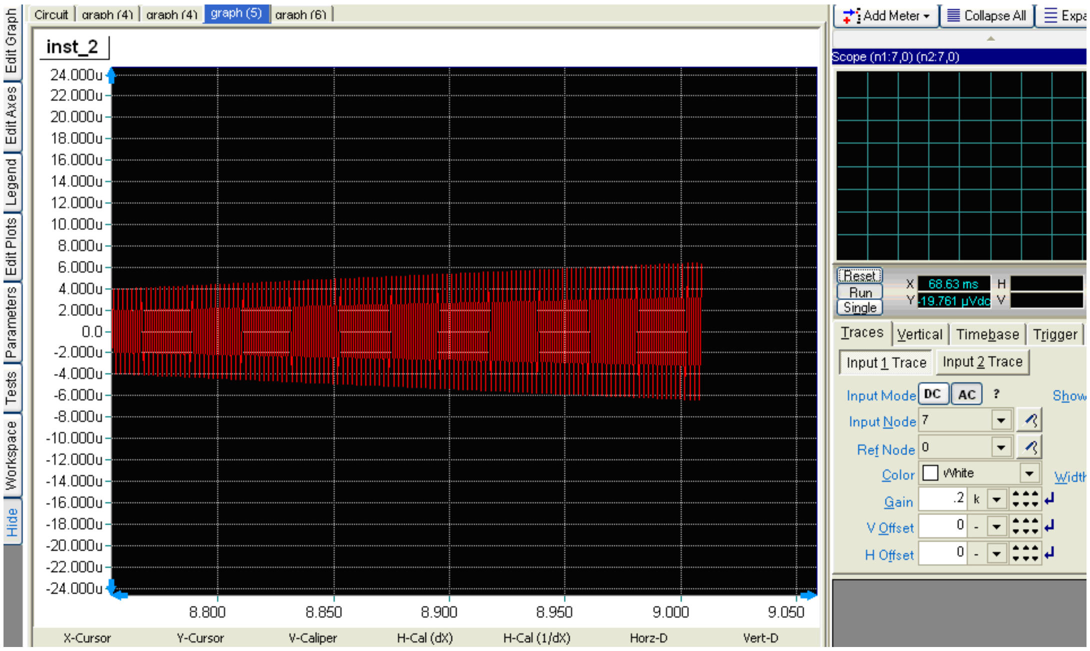


Figure 5.2: a. Simulation Result for Wein Bridge Oscillator Circuit

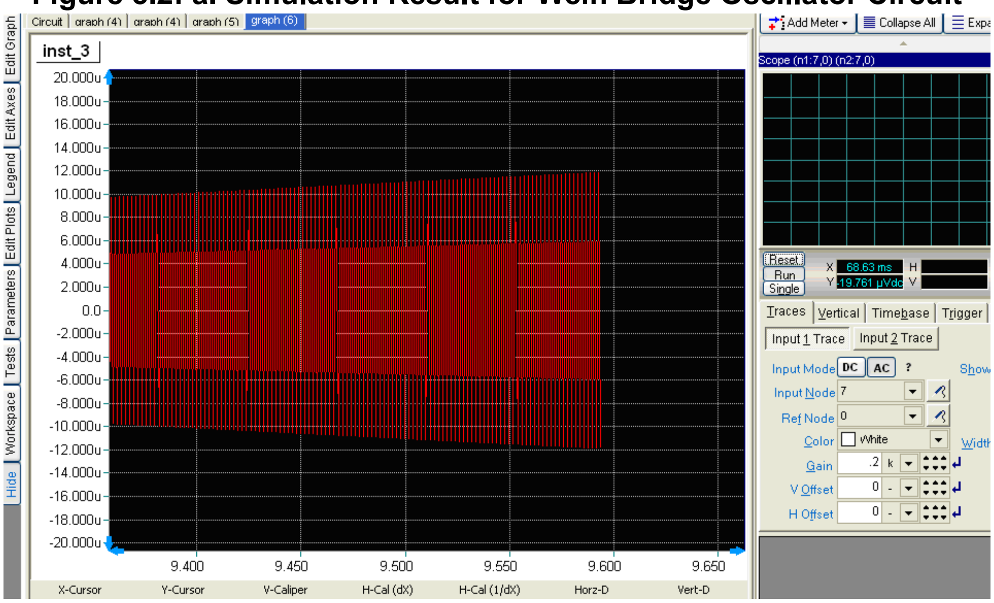


Figure 5.2: b. Simulation Result for Wein Bridge Oscillator Circuit

COMMENTS AND CONCLUSIONS

In conclusion, the Wien Bridge oscillator is easy to understand in theory, but the practicalities of the circuit can cause the designer frustration. Insertion of a digipot in the three critical areas of the circuit ensures much more stable operation and factory/user adjustability. From the simulation result we can know about the pros and cons of the oscillator.

EXPERIMENT NO.6

EXPERIMENT NO:6

NAME

RC Phase Shift Oscillator

OBJECTIVE

To Design and Simulate a RC Phase Shift Oscillator circuit using B^2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number of components required
BJT	Q2n3904	Transistor	1
RESISTOR	Res	Resistor	5
CAPACITOR	Cap	Capacitor	5
VDC	Vcc	Dc Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

An oscillator is a circuit, which generates ac output signal without giving any input ac signal. This circuit is usually applied for audio frequencies only. The basic requirement for an oscillator is positive feedback. The operation of the RC Phase Shift Oscillator can be explained as follows. The starting voltage is provided by noise, which is produced due to random motion of electrons in resistors used in the circuit. The noise voltage contains almost all the sinusoidal frequencies. This low amplitude noise voltage gets amplified and appears at the output terminals. The amplified noise drives the feedback network which is the phase shift network. Because of this the feedback voltage is maximum at a particular frequency, which in turn represents the frequency of oscillation. Furthermore, the phase shift required for positive feedback is correct at this frequency only. The voltage gain of the amplifier with positive feedback is given by

$$A_f = \frac{A}{1 - A\beta}$$

From the above equation we can see that if $A\beta = 1 \quad A_f = \infty$. The gain becomes infinity means that there is output without any input. i.e. the amplifier becomes an oscillator. This condition $A\beta = 1$ is known as the *Barkhausen criterion* of oscillation. Thus the output contains only a single sinusoidal frequency. In the beginning, as the oscillator is switched on, the loop gain *A*□ is greater than unity. The oscillations build up. Once a suitable level is reached the gain of the amplifier decreases, and the value of the loop gain decreases to unity. So the constant level oscillations are maintained. Satisfying the above conditions of oscillation the value of *R* and *C* for the phase shift network is selected such that each *RC* combination produces a phase shift of 60°. Thus the total phase shift produced by the three *RC* networks is 180°. Therefore at the specific frequency f_o the total phase shift from the base of the transistor around the circuit and back to the base is 360° thereby satisfying *Barkhausen criterion*. We select $R_1=R_2=R_3=R$ and $C_1=C_2=C_3=C$

The frequency of oscillation of RC Phase Shift Oscillator is given by

$$f_o = \frac{1}{2\pi RC\sqrt{6}}$$

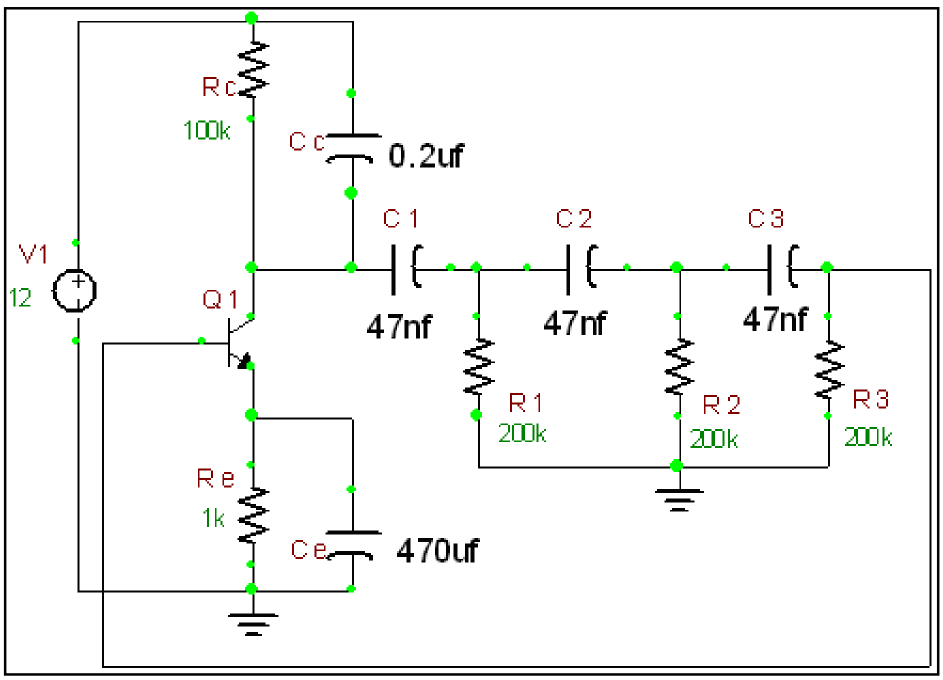


Figure 6.1: RC Phase Shift Oscillator Circuit

At this frequency, the feedback factor of the network is $\beta = \frac{1}{29}$. In order that $|A\beta| < 1$ it is required that the amplifier gain $|A| > 29$ for oscillator operation.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 6.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

The simulation result here is a continuous damping process where we can show them in 2 or 3 consecutive snaps of the simulation process. The below are the result of an oscillation due to the RC Phase Shift Oscillator.

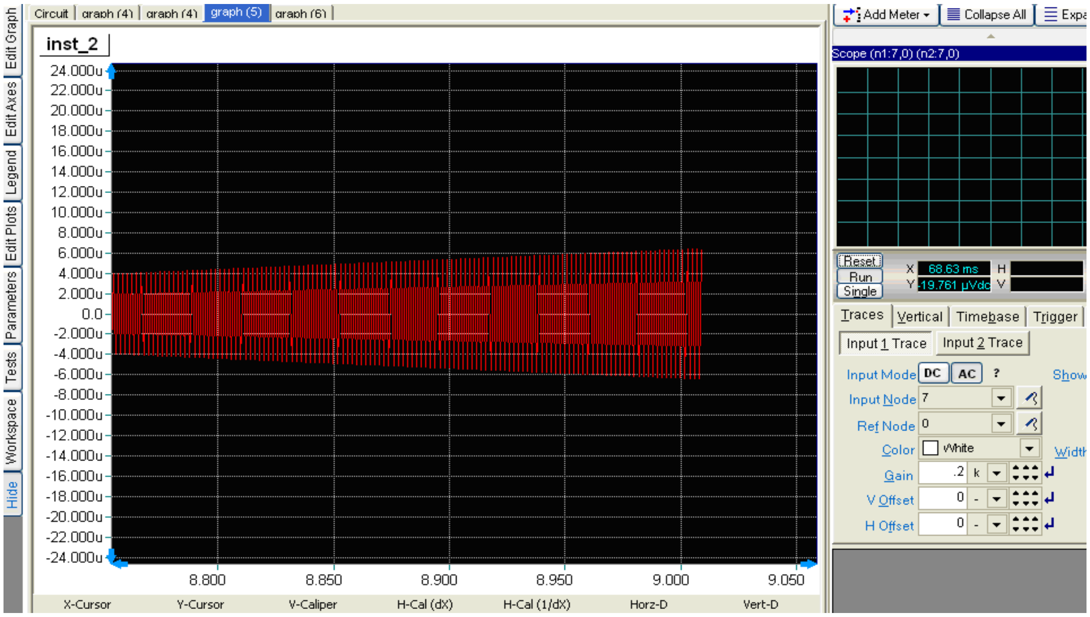


Figure 6.2: a. Simulation Result for RC Phase Shift Oscillator Circuit

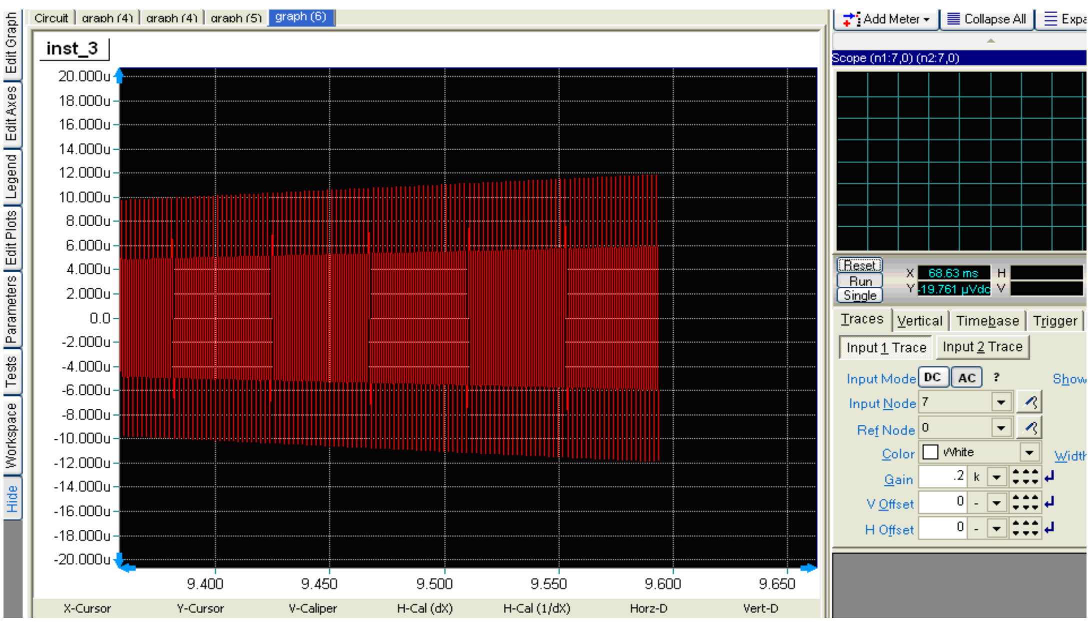


Figure 6.2: a. Simulation Result for RC Phase Shift Oscillator Circuit

COMMENTS AND CONCLUSIONS

In conclusion, the RC phase shift oscillator is easy to understand in theory. The operation of the RC Phase Shift Oscillator can be explained as follows. The starting voltage is provided by noise, which is produced due to random motion of electrons in resistors used in the circuit. From the simulation result we can know about the pros and cons of the oscillator.

EXPERIMENT NO.7

EXPERIMENT NO: 7

NAME

Class A Power Amplifier (Transformer Less)

OBJECTIVE

To design and simulate a Class A Power Amplifier circuit using B^2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number of components required
BJT	q2n3904	Transistor	1
RESISTOR	RES	Resistor	3
CAPACITOR	CAP	Capacitor	2
VGEN	VSIN	AC voltage source	1
VDC	VCC	Dc voltage source	1
BATTERY	Battery	Base bias	1
GND	GND	Ground	1

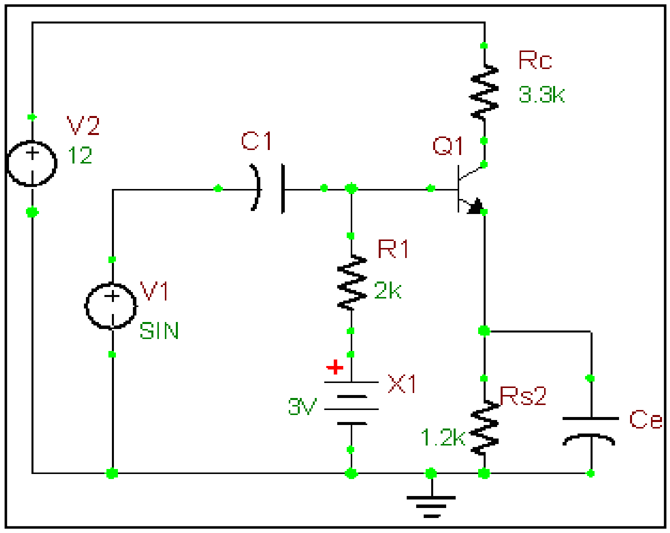
DESIGN AND THEORY

Class A amplifying devices operate over the whole of the input cycle such that the output signal is an exact scaled-up replica of the input with no clipping. Class A amplifiers are the usual means of implementing small-signal amplifiers. They are not very efficient; a theoretical maximum of 50% is obtainable with inductive output coupling and only 25% with capacitive coupling.

In a Class A circuit, the amplifying element is biased so the device is always conducting to some extent, and is operated over the most linear portion of its characteristic curve (known as its transfer characteristic or trans-conductance curve).

Because the device is always conducting, even if there is no input at all, power is drawn from the power supply. This is the chief reason for its inefficiency.

Figure 7.1: Class A Power Amplifier Circuit



If high output powers are needed from a Class A circuit, the power waste (and the accompanying heat) will become significant. For every watt delivered to the load, the amplifier itself will, at best, dissipate another watt. For large powers this means very large and expensive power supplies and heat sinking. Class A designs have largely been superseded for audio power amplifiers, though some audiophiles believe that Class A gives the best sound quality, due to it being operated in as linear a manner as possible which provides a small market for expensive high fidelity Class A amps. In addition, some aficionados prefer thermionic valve (or "tube") designs instead of transistors, for several claimed reasons:

Tubes are more commonly used in class A designs, which have an asymmetrical transfer function. This means that distortion of a sine wave creates both odd- and even-numbered harmonics. The claim is that this sounds more "musical" than the higher level of odd harmonics produced by a symmetrical push–pull amplifier.[4][5] Though good amplifier design can reduce harmonic distortion patterns to almost nothing, distortion is essential to the sound of electric guitar amplifiers, for example, and is held by recording engineers to offer more flattering microphones and to enhance "clinical-sounding" digital technology.

Valves use many more electrons at once than a transistor, and so statistical effects lead to a "smoother" approximation of the true waveform — see shot noise for more on this. Junction field-effect transistors (JFETs) have similar characteristics to valves, so these are found more often in high quality amplifiers than bipolar transistors. Historically, valve amplifiers often used a Class A power amplifier simply because valves are large and expensive; many Class A designs use only a single device.

Transistors are much cheaper, and so more elaborate designs that give greater efficiency but use more parts are still cost-effective. A classic application for a pair of class A devices is the long-tailed pair, which is exceptionally linear, and forms the basis of many more complex circuits, including many audio amplifiers and almost all op-amps. Class A amplifiers are often used in output stages of op-amps; they are sometimes used as medium-power, low-efficiency, and high-cost audio amplifiers. The power consumption is unrelated to the output power. At idle (no input), the power consumption is essentially the same as at high output volume. The result is low efficiency and high heat dissipation.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 7.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

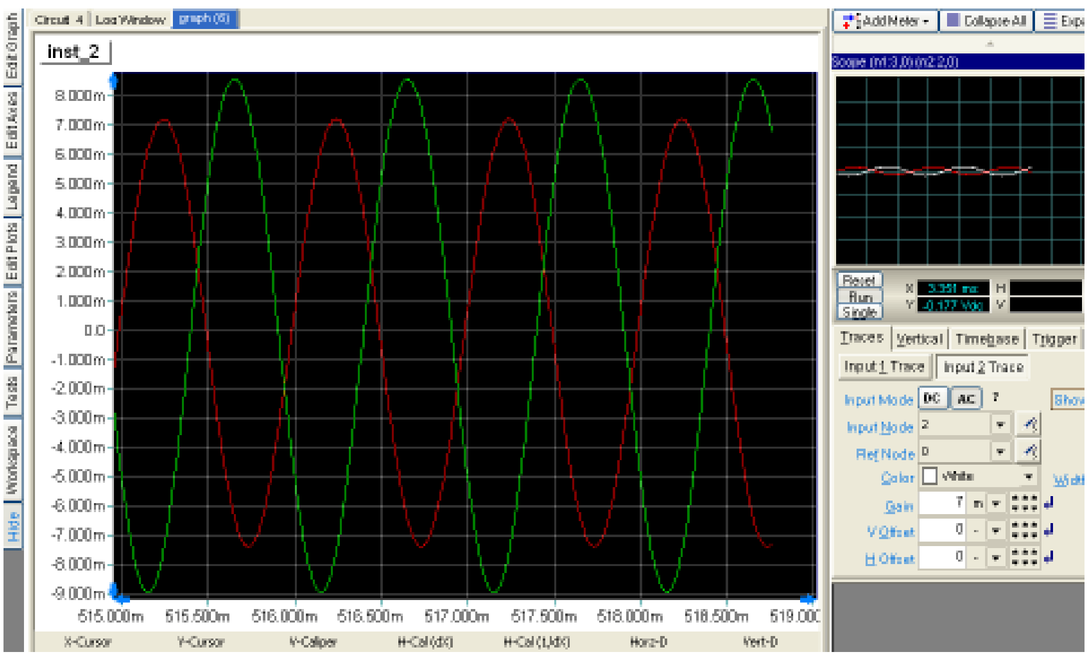


Figure 7.2: Simulation Result for Class A Power Amplifier Circuit

COMMENTS AND CONCLUSIONS

Compare your simulation results with your theoretical simulation outputs. Any significant discrepancies can be explained. Also, the purpose or function of the various components in the circuit can clearly be explained. How does this circuit generally compare to other amplifiers? The voltage gain is very high which can be seen clearly from the simulation.

EXPERIMENT NO.8

EXPERIMENT NO: 8

NAME

Class B Complementary Symmetry Amplifier

OBJECTIVE

To design and simulate a Class B Complementary Symmetry Amplifier circuit using B^2Spice.

CIRCUIT NOTE

Name	**B^2Spice Components Used**	**Description**	**Number of components required**
BJT	qbc547a (NPN), qbc557a (PNP)	Transistor	2
RESISTOR	RES	Resistor	4
CAPACITOR	CAP	Capacitor	1
VGEN	VSIN	AC voltage source	1
VDC	VCC	Dc voltage source	2
GND	GND	Ground	4

DESIGN AND THEORY

Class B amplifiers only amplify half of the input wave cycle. As such they create a large amount of distortion, but their efficiency is greatly improved and is much better than Class A. Class B has a maximum theoretical efficiency of 78.5% (i.e., π/4). This is because the amplifying element is switched off altogether half of the time, and so cannot dissipate power. A single Class B element is rarely found in practice, though it can be used in RF power amplifier where the distortion levels are less important. However Class C is more commonly used for this.

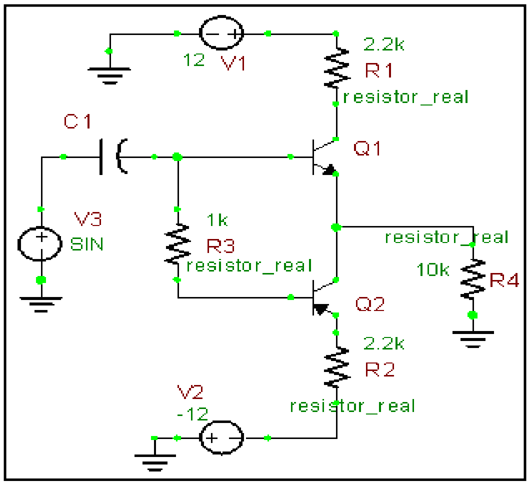


Figure 8.1: Class B Complementary Symmetry Amplifier Circuit

CLASS B AMPLIFIER

A practical circuit using Class B elements is the complementary pair or "push–pull" arrangement. Here, complementary or quasi-complementary devices are used to each amplify the opposite halves of the input signal, which is then recombined at the output. This arrangement gives excellent efficiency, but can suffer from the drawback that there is a small mismatch at the "joins" between the two halves of the signal. This is called crossover distortion. An improvement is to bias the devices so they are not completely off when they're not in use. It is typically much more efficient than class A.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 8.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

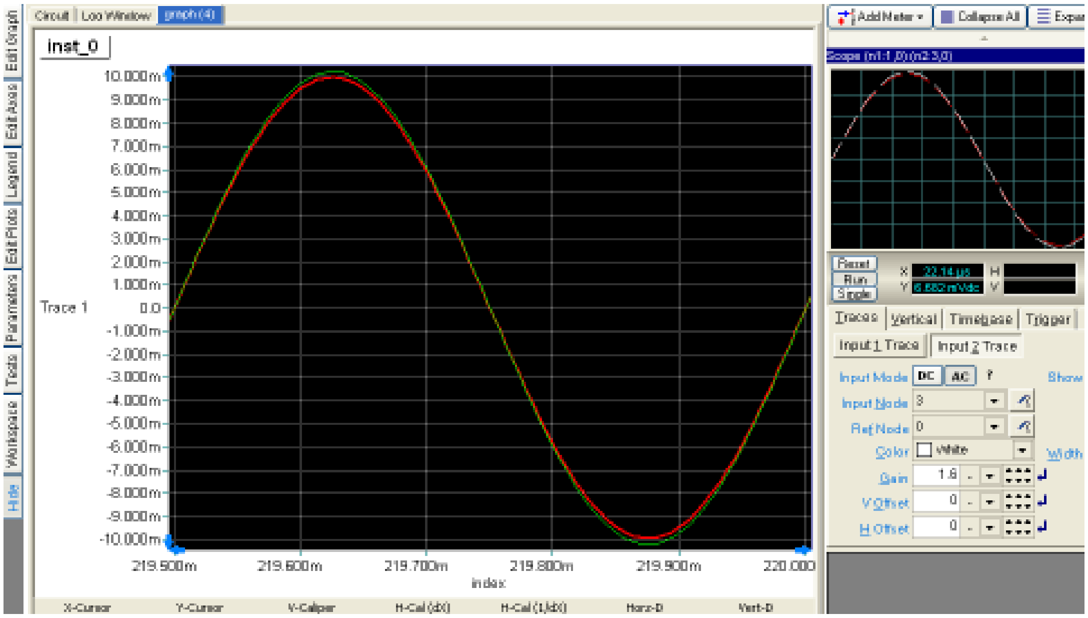


Figure 8.2: Simulation Result for Class B Complementary Symmetry Amplifier Circuit

COMMENTS AND CONCLUSIONS

Class B push–pull circuits are the most common design type found in audio power amplifiers. Class AB is widely considered a good compromise for audio amplifiers, since much of the time the music is quiet enough that the signal stays in the "class A" region, where it is amplified with good fidelity, and by definition if passing out of this region, is large enough that the distortion products typical of class B are relatively small. The crossover distortion can be reduced further by using negative feedback. Class B and AB amplifiers are sometimes used for RF linear amplifiers as well. Class B amplifiers are also favored in battery-operated devices, such as transistor radios.

EXPERIMENT NO.9

EXPERIMENT NO: 9

NAME

Common-Base Amplifier

OBJECTIVE

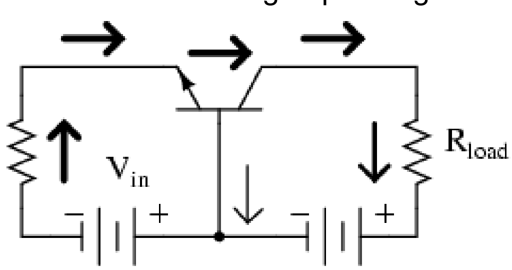
The purpose of this experiment is to investigate the operation of a common-source JFET transistor amplifier using B2Spice.

CIRCUIT NOTE

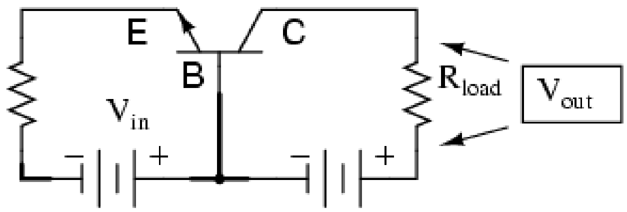
Name	B^2Spice Components Used	Description	Number Of Components Required
BJT	Qbc547a	Transistor	1
RESISTOR	R	Resistor	5
CAPACITOR	C	Capacitor	3
VGEN	Vsin	Ac Voltage Source	1
VDC	Vcc	Dc Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

The final transistor amplifier configuration we need to study is the *common-base*. This configuration is more complex than the other two, i.e. common-emitter and common collector and is less common due to its strange operating characteristics.



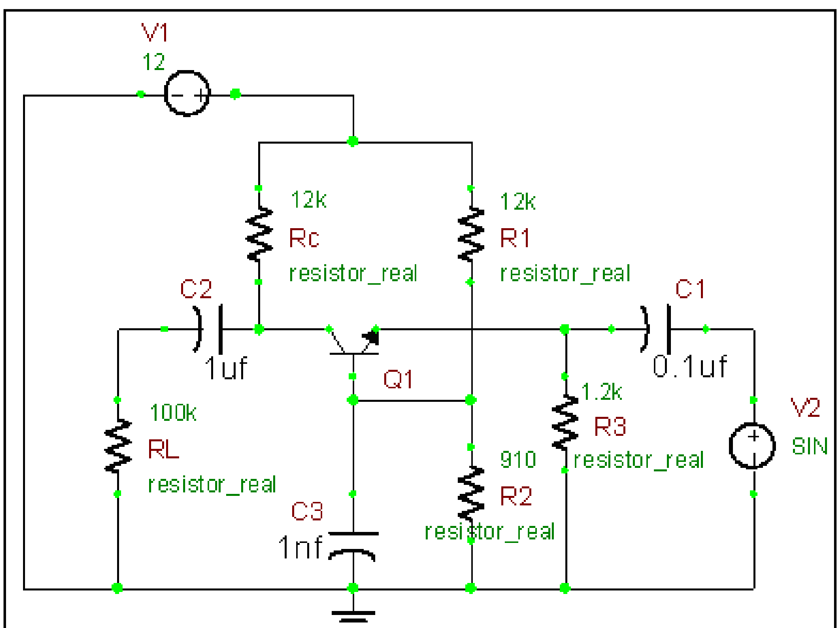
It is called the common-base configuration because (DC power source aside), the signal source and the load share the base of the transistor as a common connection point:



Perhaps the most striking characteristic of this configuration is that the input signal source must carry the full emitter current of the transistor, as indicated by the heavy arrows in the first illustration. As we know, the emitter current is greater than any other current in the transistor, being the sum of base and collector currents. In the last two amplifier configurations, the signal source was connected to the base lead of the transistor, thus handling the least current possible.

Because the input current exceeds all other currents in the circuit, including the output current, the current gain of this amplifier is actually less than 1 (notice how RL is connected to the collector, thus carrying slightly less current than the signal source). In other words, it attenuates current rather than amplifying it. With common-emitter and common-collector amplifier configurations, the transistor parameter most closely associated with gain was β. In the common-base circuit, we follow another basic transistor parameter: the ratio between collector current and emitter current, which is a fraction always less than 1. This fractional value for any transistor is called the alpha ratio, or α ratio.

Since it obviously can't boost signal current, it only seems reasonable to expect it to



boost signal voltage. A B²Spice simulation will vindicate that assumption:

Figure 9.1: Common-Base Amplifier Circuit

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 9.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal; for example, by temporarily removing CS, you can see that the output signal dramatically decreases. By changing the source and drain resistors, it may be possible to further optimize the amplifier for a larger peak output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

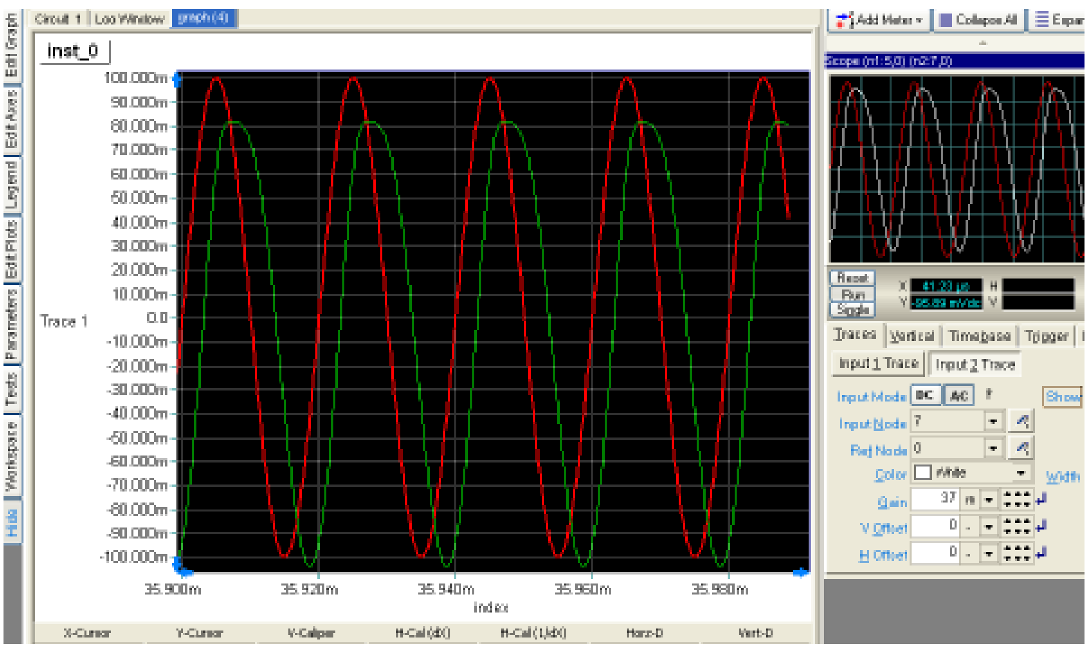


Figure 9.2: Simulation Result for Common-Base Amplifier Circuit

COMMENTS AND CONCLUSIONS

Notice how in this simulation represents a rather large voltage gain a gain ratio of 37.5, or 31.48 dB. Notice also how the output voltage (measured across R_{load}) actually exceeds the power supply (15 volts) at saturation, due to the series-aiding effect of the input voltage source. Compare your results with your PRE-LAB calculations. Any significant discrepancies can be explained. Also, the purpose or function of the various components in the circuit can clearly be explained; for example, how does this circuit generally compare to a common-gate amplifier?

EXPERIMENT NO.10

EXPERIMENT NO: 10

NAME

Biasing Of Active Circuits

OBJECTIVE

To Design and Simulate different types of biasing circuits for BJT and JFET using B2Spice

DESIGN & THEORY

Biasing in electronics is the method of establishing predetermined voltages and/or currents at various points of a circuit to set an appropriate operating point.

The operating point of a device, also known as bias point or quiescent point (or simply Q-point), is the DC voltage and/or current which, when applied to a device, causes it to operate in a certain desired fashion. The term is normally used in connection with devices such as transistors and diodes which are used in amplification or rectification. Linear circuits involving transistors typically require specific DC voltages and currents to operate correctly, which can be achieved using a biasing circuit.

REQUIREMENTS OF BIASING CIRCUIT

The operating point of a device, also known as bias point or quiescent point (or simply Q-point), is the DC voltage and/or current which, when applied to a device, causes it to operate in a certain desired fashion.

For analog circuit operation, the Q-point is placed so the transistor stays in active mode (does not shift to operation in the saturation region or cut-off region) when input is applied. For digital operation, the Q-point is placed so the transistor does the contrary - switches from "on" to "off" state. Often, Q-point is established near the center of active region of transistor characteristic to allow similar signal swings in positive and negative directions.

Q-point should be stable. In particular, it should be insensitive to variations in transistor parameters (for example, should not shift if transistor is replaced by another of the same type), variations in temperature, variations in power supply voltage and so forth.

The circuit must be practical: easily implemented and cost-effective.

DISCRETE BIPOLAR JUNCTION TRANSISTOR BIASING

Types of bias circuit

The following discussion treats five common biasing circuits used with bipolar transistors:

- Collector-to-base bias
- Fixed bias with emitter resistor
- Voltage divider bias

COLLECTOR-TO-BASE BIAS

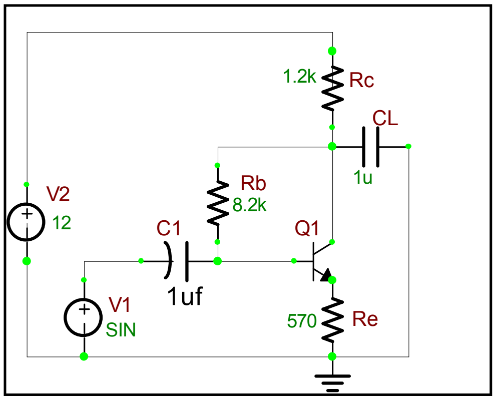


Figure 10.1: Collector-to-base bias

In this form of biasing, the base resistor R_B is connected to the collector instead of connecting it to the battery V_{CC}. That means this circuit employs negative feedback to stabilize the operating point.

From Kirchhoff's voltage law, the voltage across the base resistor is

$V_{Rb} = V_{CC} - (I_C + I_b)\, RC - V_{be}$.

From Ohm's law, the base current is

$I_b = V_{Rb} / R_b$.

The way feedback controls the bias point is as follows. If V_{be} is held constant and temperature increases, collector current increases. However, a larger I_C causes the voltage drop across resistor R_C to increase, which in turn reduces the voltage V_{Rb} across the base resistor. A lower base-resistor voltage drop reduces the base current, which results in fewer collectors current, so increase in collector current with temperature is opposed, and operating point is kept stable.

For the given circuit,

$I_B = (V_{CC} - V_{be}) / (R_B+\beta R_C)$.

Merits

Circuit stabilizes the operating point against variations in temperature and β (i.e. replacement of transistor)

Demerits:

In this circuit, to keep I_C independent of β the following condition must be met:

$$I_C = \beta I_B = \frac{\beta(V_{CC} - V_{be})}{R_B + \beta R_C} \approx \frac{(V_{CC} - V_{be})}{R_C}$$

This is approximately the case if

$\beta R_C >> R_B$.

As β-value is fixed for a given transistor, this relation can be satisfied either by keeping R_C fairly large, or making R_B very low.

If R_C is of large value, high V_{CC} is necessary. This increases cost as well as precautions necessary while handling.

If R_B is low, the reverse bias of the collector-base is small, which limits the range of collector voltage swing that leaves the transistor in active mode.

The resistor R_B causes an ac feedback, reducing the voltage gain of the amplifier. This undesirable effect is a trade-off for greater Q-point stability.

Usage: The feedback also decreases the input impedance of the amplifier as seen from the base, which can be advantageous. Due to the gain reduction from feedback, this biasing form is used only when the trade-off for stability is warranted.

FIXED BIAS WITH EMITTER RESISTOR

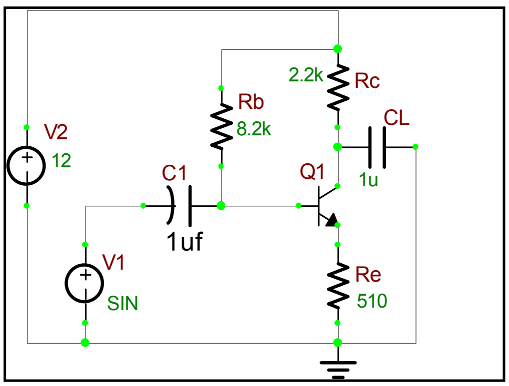


Figure 10.2: Fixed bias with emitter resistor

The fixed bias circuit is modified by attaching an external resistor to the emitter. This resistor introduces negative feedback that stabilizes the Q-point. From Kirchhoff's voltage law, the voltage across the base resistor is

$V_{Rb} = V_{CC} - I_eR_e - V_{be}$.

From Ohm's law, the base current is

$I_b = V_{Rb} / R_b$.

The way feedback controls the bias point is as follows. If V_{be} is held constant and temperature increases, emitter current increases. However, a larger I_e increases the emitter voltage $V_e = I_eR_e$, which in turn reduces the voltage V_{Rb} across the base resistor. A lower base-resistor voltage drop reduces the base current, which results in less collector current because $I_c = ß\ I_B$. Collector current and emitter current are related by $I_c = \alpha\ I_e$ with $\alpha \approx 1$, so increase in emitter current with temperature is opposed, and operating point is kept stable.

Similarly, if the transistor is replaced by another, there may be a change in I_C (corresponding to change in β-value, for example). By similar process as above, the change is negated and operating point kept stable.

For the given circuit,

$I_B = (V_{CC} - V_{be}) / (R_B + (\beta+1)\ R_E)$.

Merits

The circuit has the tendency to stabilize operating point against changes in temperature and β-value.

Demerits

In this circuit, to keep I_C independent of β the following condition must be met:

$$I_C = \beta I_B = \frac{\beta(V_{CC} - V_{be})}{R_B + (\beta+1)R_E} \approx \frac{(V_{CC} - V_{be})}{R_E}$$

This is approximately the case if

$(\beta + 1) R_E >> R_B$.

As β-value is fixed for a given transistor, this relation can be satisfied either by keeping R_E very large, or making R_B very low.

If R_E is of large value, high V_{CC} is necessary. This increases cost as well as precautions necessary while handling.

If R_B is low, a separate low voltage supply should be used in the base circuit. Using two supplies of different voltages is impractical.

In addition to the above, R_E causes ac feedback which reduces the voltage gain of the amplifier.

Usage

The feedback also increases the input impedance of the amplifier as seen from the base, which can be advantageous. Due to the above disadvantages, this type of biasing circuit is used only with careful consideration of the trade-offs involved.

VOLTAGE DIVIDER BIAS

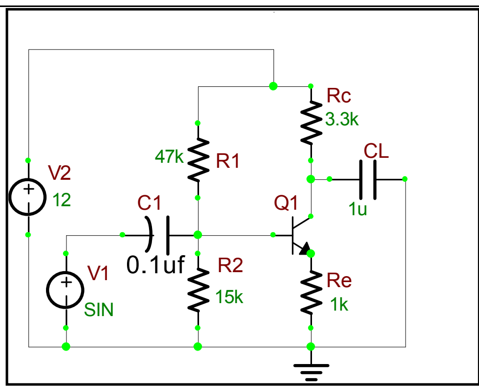


Figure 10.3: Voltage divider bias

The voltage divider is formed using external resistors R_1 and R_2. The voltage across R_2 forward biases the emitter junction. By proper selection of resistors R_1 and R_2, the operating point of the transistor can be made independent of β. In this circuit, the voltage divider holds the base voltage fixed independent of base current provided the divider current is large compared to the base current. However, even with a fixed base voltage, collector current varies with temperature (for example) so an emitter resistor is added to stabilize the Q-point, similar to the above circuits with emitter resistor.

In this circuit the base voltage is given by:

$V_B =$Voltage across R_2 $\quad = V_{cc}\dfrac{R_2}{(R_1+R_2)} - I_B\dfrac{R_1R_2}{(R_1+R_2)}$

$\approx V_{cc}\frac{R_2}{(R_1 + R_2)}$ Provided $I_B << I_2 = V_B/R_2$.

Also $V_B = V_{be} + I_E R_E$

For the given circuit,

$$I_B = \frac{\frac{V_{CC}}{1+R_1/R_2} - V_{be}}{(\beta+1)R_E + R_1 \parallel R_2}.$$

Merits

Unlike above circuits, only one dc supply is necessary.

Operating point is almost independent of β variation.

Operating point stabilized against shift in temperature.

Demerits

In this circuit, to keep I_C independent of β the following condition must be met:

$$I_C = \beta I_B = \beta\frac{\frac{V_{CC}}{1+R_1/R_2} - V_{be}}{(\beta+1)R_E + R_1 \parallel R_2} \approx \frac{\frac{V_{CC}}{1+R_1/R_2} - V_{be}}{R_E},$$

This is approximately the case if

$(\beta+1)R_E >> R_1 \parallel R_2$

Where $R_1 \parallel R_2$ denotes the equivalent resistance of R_1 and R_2 connected in parallel.

As β-value is fixed for a given transistor, this relation can be satisfied either by keeping R_E fairly large or making $R_1||R_2$ very low.

If R_E is of large value, high V_{CC} is necessary. This increases cost as well as precautions necessary while handling.

If $R_1 \parallel R_2$ is low, either R_1 is low, or R_2 is low, or both are low. A low R_1 raises V_B closer to V_C, reducing the available swing in collector voltage, and limiting how large R_C can be made without driving the transistor out of active mode. A low R_2 lowers V_{be}, reducing the allowed collector current. Lowering both resistor values draws more current from the power supply and lowers the input resistance of the amplifier as seen from the base.

AC as well as DC feedback is caused by R_E, which reduces the AC voltage gain of the amplifier. A method to avoid AC feedback while retaining DC feedback is discussed below.

Usage

The circuit's stability and merits as above make it widely used for linear circuits.

DISCRETE JUNCTION FIELD EFFECT TRANSISTOR BIASING

Fixed bias with source resistor

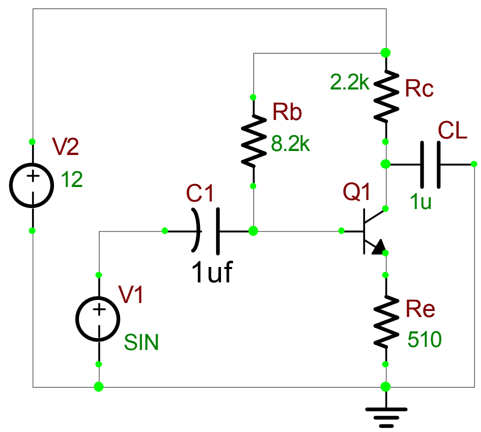


Figure 10.4: Fixed bias with source resistor

The fixed bias circuit is modified by attaching an external resistor to the source. This resistor introduces negative feedback that stabilizes the Q-point. From Kirchhoff's voltage law, the voltage across the gate resistor is

$V_{Rg} = V_{CC} - I_sR_s - V_{gs}$.

From Ohm's law, the base current is

$I_g = V_{Rg} / R_g$.

The way feedback controls the bias point is as follows. If V_{gs} is held constant and temperature increases, source current increases. However, a larger I_s increases the source voltage $V_s = I_sR_s$, which in turn reduces the voltage V_{Rg} across the gate resistor. A lower gate-resistor voltage drop reduces the gate current, which results in less drain current. Drain current and source current are related by $I_d \approx I_s$, so increase in source current with temperature is opposed, and operating point is kept stable.

Similarly, if the transistor is replaced by another, there may be a change in I_D. By similar process as above, the change is negated and operating point kept stable.

Merits

The circuit has the tendency to stabilize operating point against changes in temperature and β-value.

Demerits

In this circuit, to keep I_C independent of β the following condition must be met:

$$I_C = \beta I_B = \frac{\beta(V_{CC} - V_{be})}{R_B + (\beta + 1)R_E} \approx \frac{(V_{CC} - V_{be})}{R_E}$$

This is approximately the case if

$(\beta + 1) R_E >> R_B$.

As β-value is fixed for a given transistor, this relation can be satisfied either by keeping R_E very large or making R_B very low.

If R_E is of large value, high V_{CC} is necessary. This increases cost as well as precautions necessary while handling.

If R_G is low, a separate low voltage supply should be used in the gate circuit. Using two supplies of different voltages is impractical.

In addition to the above, R_E causes ac feedback which reduces the voltage gain of the amplifier

Usage

The feedback also increases the input impedance of the amplifier as seen from the gate, which can be advantageous. Due to the above disadvantages, this type of biasing circuit is used only with careful consideration of the trade-offs involved.

VOLTAGE DIVIDER BIAS

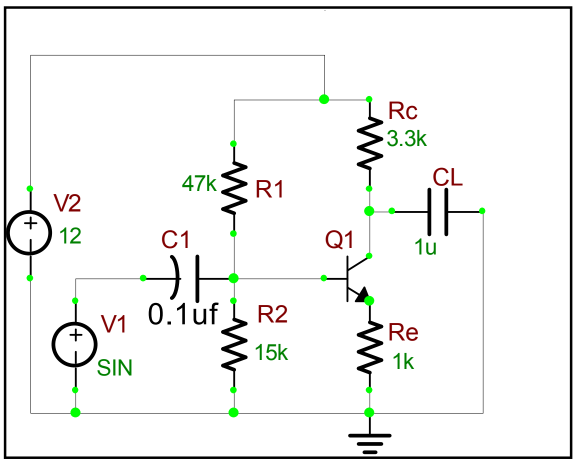


Figure 10.5: Voltage divider bias

The voltage divider is formed using external resistors R_1 and R_2. The voltage across R_2 forward biases the source junction. By proper selection of resistors R_1 and R_2, the operating point of the transistor can be made independent of β. In this circuit, the voltage divider holds the gate voltage fixed independent of gate current provided the divider current is large compared to the gate current. However, even with a fixed gate voltage, drain current varies with temperature (for example) so a source resistor is added to stabilize the Q-point, similar to the above circuits with source resistor.

In this circuit the gate voltage is given by:

V_B =Voltage across R_2 $= V_{cc}\frac{R_2}{(R_1+R_2)} - I_B\frac{R_1R_2}{(R_1+R_2)}$

$\approx V_{cc}\frac{R_2}{(R_1+R_2)}$ Provided $I_B << I_2 = V_B/R_2$.

Also $V_B = V_{be} + I_E R_E$

For the given circuit,

$$I_B = \frac{\frac{V_{CC}}{1+R_1/R_2} - V_{be}}{(\beta+1)R_E + R_1 \parallel R_2}.$$

Merits

- Unlike above circuits, only one dc supply is necessary.
- Operating point is almost independent of β variation.
- Operating point stabilized against shift in temperature.

Demerits

In this circuit, to keep I_C independent of β the following condition must be met:

$$I_C = \beta I_B = \beta \frac{\frac{V_{CC}}{1+R_1/R_2} - V_{be}}{(\beta+1)R_E + R_1 \parallel R_2} \approx \frac{\frac{V_{CC}}{1+R_1/R_2} - V_{be}}{R_E},$$

This is approximately the case if

$$(\beta+1)R_E >> R_1 \parallel R_2$$

Where $R_1 \parallel R_2$ denotes the equivalent resistance of R_1 and R_2 connected in parallel.

As β-value is fixed for a given transistor, this relation can be satisfied either by keeping R_E fairly large or making $R_1||R_2$ very low.

If R_E is of large value, high V_{CC} is necessary. This increases cost as well as precautions necessary while handling.

If $R_1 \parallel R_2$ is low, either R_1 is low, or R_2 is low, or both are low. A low R_1 raises V_B closer to V_C, reducing the available swing in drain voltage, and limiting how large R_C can be made without driving the transistor out of active mode. A low R_2 lowers V_{be}, reducing the allowed drain current. Lowering both resistor values draws more current from the power supply and lowers the input resistance of the amplifier as seen from the gate.

AC as well as DC feedback is caused by R_E, which reduces the AC voltage gain of the amplifier. A method to avoid AC feedback while retaining DC feedback is discussed below.

Usage

The circuit's stability and merits as above make it widely used for linear circuits.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuits shown for all types of biasing using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal; for example, by temporarily removing CS, you can see that the output signal dramatically decreases. By changing the source and drain resistors, it may be possible to further optimize the amplifier for a larger peak output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

Here for this experiment in order to perform the DC analyses we need to carry out few more steps after Step 9. Refer to the steps mentioned in Experiment 2 from Step 10 to Step 11 to simulate for the DC Analyses and analyze the values to be plotted in order to find out ‘Q point’.

COMMENTS AND CONCLUSION

The circuit's stability and merits as above make it widely used for linear circuits. Hence the different types of biasing circuits are studied from the above experiment for both BJT and JFET.

EXPERIMENT NO.11

EXPERIMENT NO: 11

NAME

Darlington Pair

OBJECTIVE

The purpose of this experiment is to investigate the operation of a common-source JFET transistor amplifier using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number of components required
JFET	Qbc547a	Transistor	2
RESISTOR	R	Resistor	5
CAPACITOR	C	Capacitor	2
VGEN	Vsin	Ac Voltage Source	1
VDC	Vcc	Dc Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

In electronics, the Darlington transistor (often called a Darlington pair) is a semiconductor device which combines two bipolar junction transistors in a single device so that the current amplified by the first is amplified further by the second. This gives a high current gain (written β or hFE), and takes less space than two discrete transistors in the same configuration. Integrated packaged devices are available, but it is still common also to use two separate transistors.

The Darlington configuration was invented by Bell Laboratories engineer Sidney Darlington in 1953. He patented the idea of having two or three transistors on a single chip, but not that of an arbitrary number (which might have covered all modern integrated circuits). Figure 11.1 shows the B^2Spice circuit model of a Darlington Pair.

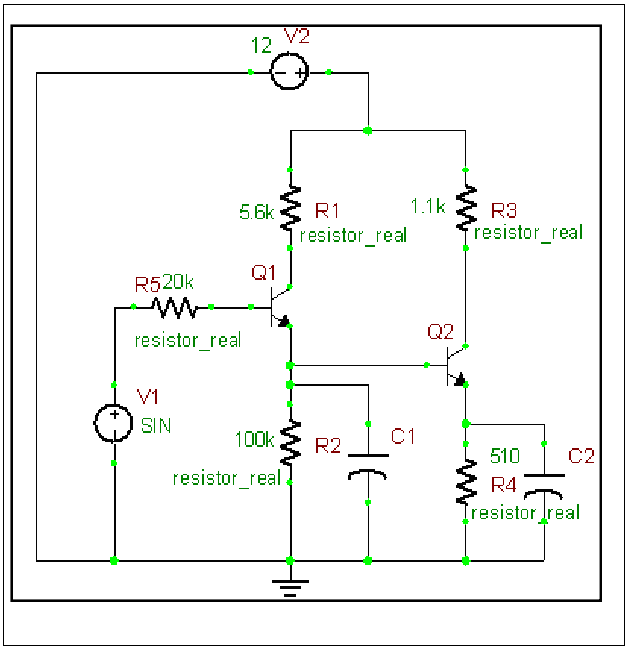


Figure 11.1: Darlington Pair

BEHAVIOUR

A Darlington pair behaves like a single transistor with a high current gain (the product of the gains of the two transistors):

$$\beta_{\text{Darlington}} = \beta_1 \cdot \beta_2$$

A typical modern device has a current gain of 1000 or more, so that only a tiny base current is needed to make the pair switch on. Integrated devices have three leads (B, C and E), broadly equivalent to those of a standard transistor.

The base-emitter voltage is also higher. It is the sum of both base-emitter voltages:

$V_{BE} = V_{BE1} + V_{BE2}$

Thus, for Si based technology, there must be about 0.7 V across both base-emitter junctions (connected in series in the device), so that we need about 1.4 V in total to turn on the device. The saturation voltage of a Darlington pair is about 0.7 V, which can cause substantial power dissipation. Another drawback is a reduction in switching speed, because the first transistor cannot actively inhibit the base current of the second, which makes the device slow to switch off. To alleviate this, the second transistor often has a base resistor of a few hundred ohms. The Darlington has more phase shift at high frequencies than a single transistor and hence can more easily become unstable with feedback.

Darlington pairs can be made from two discrete transistors; Q_1 (the left-hand transistor in the diagram) can be a low power type, but normally Q_2 (on the right) will need to be high power. The maximum collector current I_C (max) of the pair is that of Q_2.

A Darlington pair can be sensitive enough to respond to the current passed by skin contact even at safe voltages. Thus, it can form the input stage of a touch-sensitive switch.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 11.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

In step 7 instead of adding an oscilloscope add a frequency plotter.

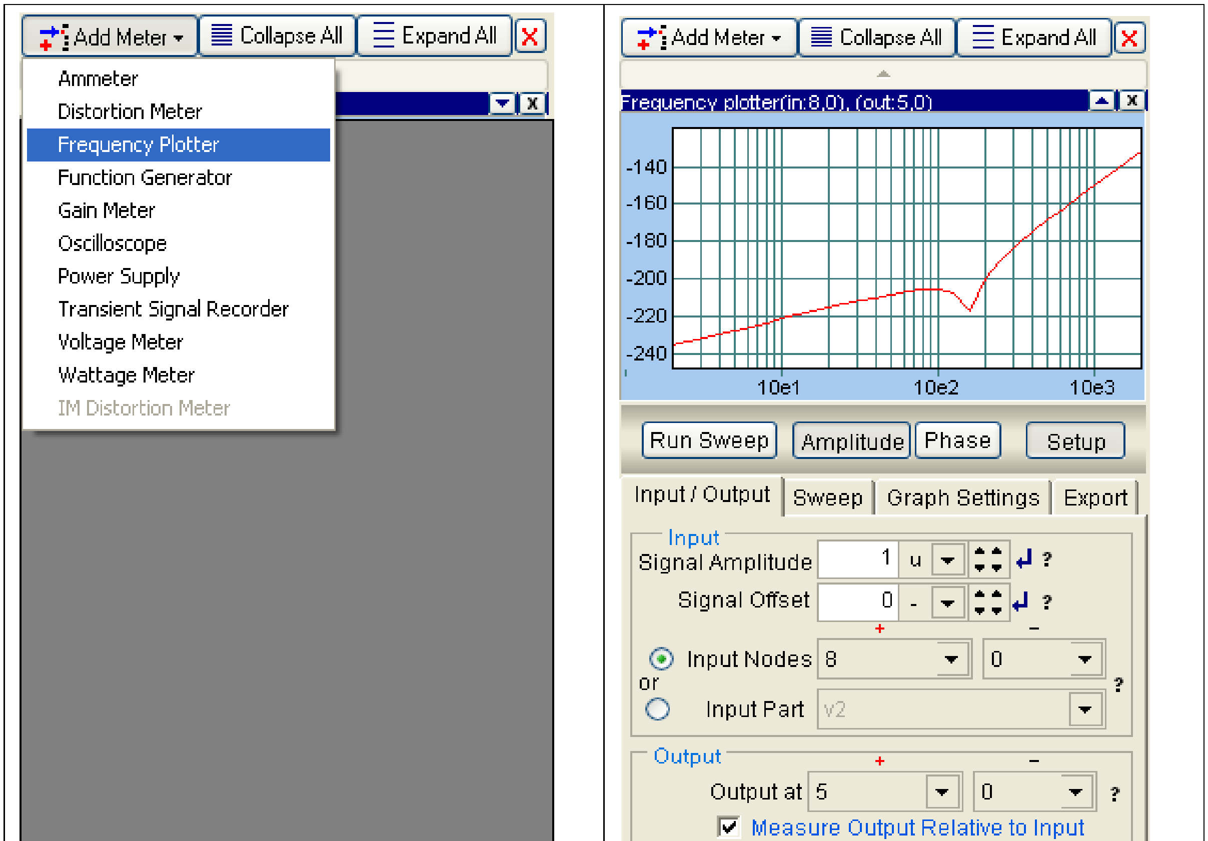


Figure 11.2: Adding Frequency Plotter

Click on Run Sweep to get the Frequency response of Darlington Pair.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B2Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

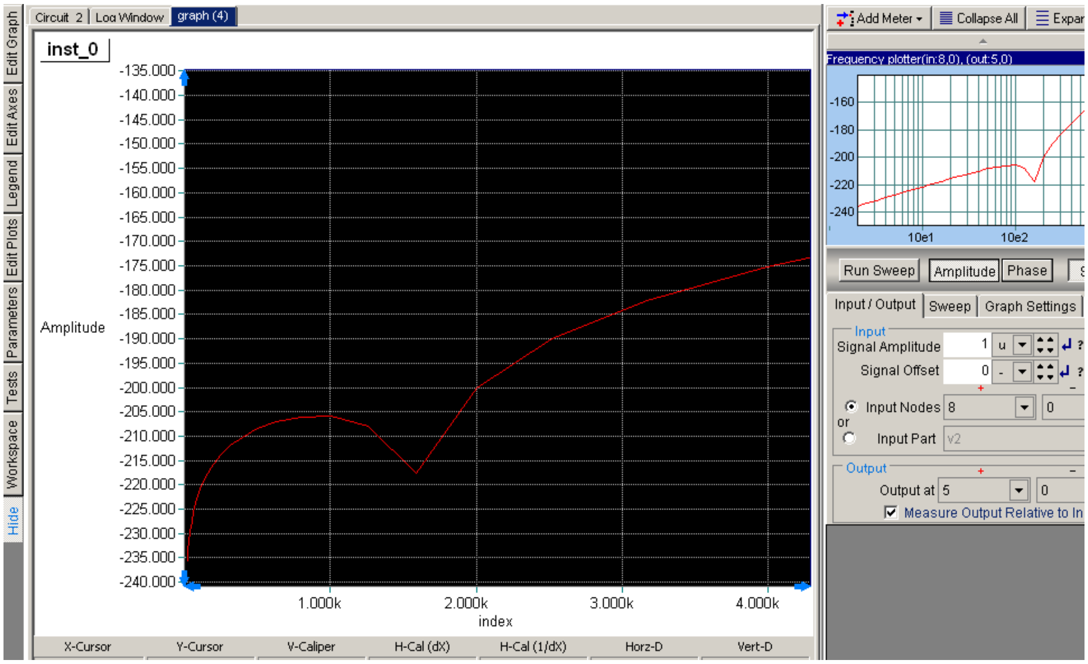


Figure 11.3: Frequency Response of a Darlington Pair

COMMENTS AND CONCLUSIONS

Any significant discrepancies can be explained. Also, the purpose or function of the various components in the circuit can clearly be explained. The frequency response for the Darlington circuit is found in the simulation. The voltage gain of the Darlington pair is close to unity.

EXPERIMENT NO.12

EXPERIMENT NO: 12

NAME

Half Wave Rectifier

OBJECTIVE

To design and simulate a Half Wave Rectifier circuit using B^2Spice.

CIRCUIT NOTE

Name	**B^2Spice Components Used**	**Description**	**Number of components required**
DIODE	Bal47	D	1
RESISTOR	R	Resistor	1
VIN	Vsin	Ac Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

A rectifier is an electrical device that converts alternating current (AC) to direct current (DC), a process known as rectification. Rectifiers have many uses including as components of power supplies and as detectors of radio signals. Rectifiers may be made of solid state diodes, vacuum tube diodes, mercury arc valves, and other components.

When only one diode is used to rectify AC (by blocking the negative or positive portion of the waveform), the difference between the term diode and the term rectifier is merely one of usage, i.e., the term rectifier describes a diode that is being used to convert AC to DC. Almost all rectifiers comprise a number of diodes in a specific arrangement for more efficiently converting AC to DC than is possible with only one diode. Before the development of silicon semiconductor rectifiers, vacuum tube diodes and copper (I) oxide or selenium rectifier stacks were used.

- The Half wave rectifier is a circuit, which converts an ac voltage to dc voltage.
- In the Half wave rectifier circuit shown above the transformer serves two purposes.
- It can be used to obtain the desired level of dc voltage (using step up or step down transformers).
- It provides isolation from the power line.
- The primary of the transformer is connected to ac supply. This induces an ac voltage across the secondary of the transformer.

During the positive half cycle of the input voltage the polarity of the voltage across the secondary forward biases the diode. As a result a current I_L flows through the load resistor, R_L. The forward biased diode offers a very low resistance and hence the voltage drop across it is very small. Thus the voltage appearing across the load is practically the same as the input voltage at every instant.

During the negative half cycle of the input voltage the polarity of the secondary voltage gets reversed. As a result, the diode is reverse biased. Practically no current flows through the circuit and almost no voltage is developed across the resistor. All input voltage appears across the diode itself.

Hence we conclude that when the input voltage is going through its positive half cycle, output voltage is almost the same as the input voltage and during the negative half cycle no voltage is available across the load. This explains the unidirectional pulsating dc waveform obtained as output. The process of removing one half the input signals to establish a dc level is aptly called half wave rectification.

When the input voltage reaches its maximum value Vm during the negative half cycle the voltage across the diode is also maximum. This maximum voltage is known as the peak inverse voltage. Thus for a half wave rectifier Peak Inverse Voltage,

$$PIV = V_m$$

Let Vi be the voltage to the primary of the transformer. Vi is given by

$$V_i = V_m \sin \omega t, V_m >> V_r \rightarrow (1)$$

Where Vr is the cut-in voltage of the diode.

Ripple factor is defined as the ratio of rms value of ac component to the dc component in the output.

Ripple factor $r = \dfrac{RMS\ value of\ the\ ac\ compo nen.}{dc\ value of\ the\ component}$

$$\gamma = \frac{Vr_{rms}}{V_{dc}} \rightarrow (2)$$

$$V\gamma_{rms} = \sqrt{V_{rms}{}^2 - V_{dc}{}^2} \rightarrow (3)$$

$$r = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1} \rightarrow (4)$$

Vav the average or the dc content of the voltage across the load is given by

$$V_{av} = V_{dc} = \frac{1}{2\pi}\left[\int_0^{\pi} V_m \sin \omega t\ d(\omega t) + \int_{\pi}^{2\pi} 0.\ d(\omega t)\right] \rightarrow (5)$$

$$= \frac{V_m}{2\pi}\left[-\cos \omega t\right]_0^{\pi} = \frac{V_m}{\pi} \rightarrow (6)$$

$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{V_m}{\pi R_L} = \frac{I_m}{\pi} \rightarrow (7)$$

RMS voltage at the load resistance can be calculated as

$$V_{rms} = \left[\frac{1}{2\pi}\int_0^{\pi} V_m{}^2 \sin^2 \omega t\ d(\omega t)\right]^{1/2} \rightarrow (8)$$

$$= V_m\left[\frac{1}{4\pi}\int_0^{\pi}(1 - \cos 2\,\omega t)\ d(\omega t)\right]^{1/2} = \frac{V_m}{2} \rightarrow (9)$$

Ripple Factor $r = \sqrt{\left(\frac{V_m/2}{V_m/\pi}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2}\right)^2} = \underline{\underline{1.21}} \rightarrow (10)$

Efficiency, □ is the ratio of the dc output power to ac input power

Thus $\eta = \frac{dc\, output power}{ac\, input power} = \frac{P_{dc}}{P_{ac}} \rightarrow (11)$

$$\frac{V_{dc}^2 / R_L}{V_{rms}^2 / R_L} = \frac{\left[V_m/\pi\right]^2}{\left[V_m/2\right]^2} = 4/\pi^2 = 0.406 = \underline{\underline{40.6\%}} \rightarrow (12)$$

Transformer Utilization Factor, TUF can be used to determine the rating of a transformer secondary.

$$TUF = \frac{P_{dc}}{P_{ac\ \ rated}} \rightarrow (13)$$

In half wave rectifier the rated voltage of the transformer secondary is $\frac{V_m}{\sqrt{2}}$

But actually the RMS current flowing through the winding is only $\frac{I_m}{2}$.

$$TUF = \frac{\left(\frac{I_m^2}{\pi^2} R_L\right)}{\left(\frac{V_m}{\sqrt{2}} \frac{I_m}{2}\right)} = \frac{\left(\frac{V_m^2}{\pi^2} \frac{1}{R_L}\right)}{\left(\frac{V_m}{\sqrt{2}} \frac{V_m}{2R_L}\right)} = \frac{2\sqrt{2}}{\pi^2} = \underline{\underline{0.287}}$$

Form factor is given by, $Form\ factor = \frac{rms\, value}{average value}$

$$= \frac{\left(V_m/2\right)}{\left(V_m/\pi\right)} = \frac{\pi}{2} = \underline{\underline{1.57}}$$

Peak factor is given by, $Peak\ \ factor = \frac{Peak\ \ value}{rms\ \ value} = \frac{V_m}{\left(V_m/2\right)} = \underline{\underline{2}}$

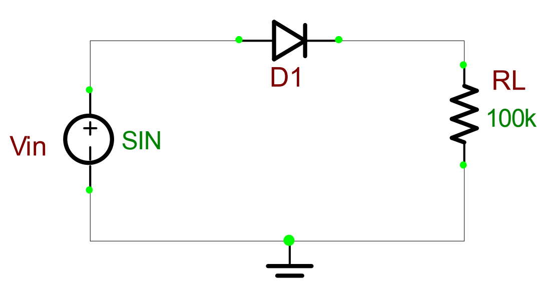


Figure 12.1: Half Wave Rectifier

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 12.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal; for example, by temporarily removing CS, you can see that the output signal dramatically decreases. By changing the source and drain resistors, it may be possible to further optimize the amplifier for a larger peak output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process. The output waveform may be observed in the Oscilloscope.

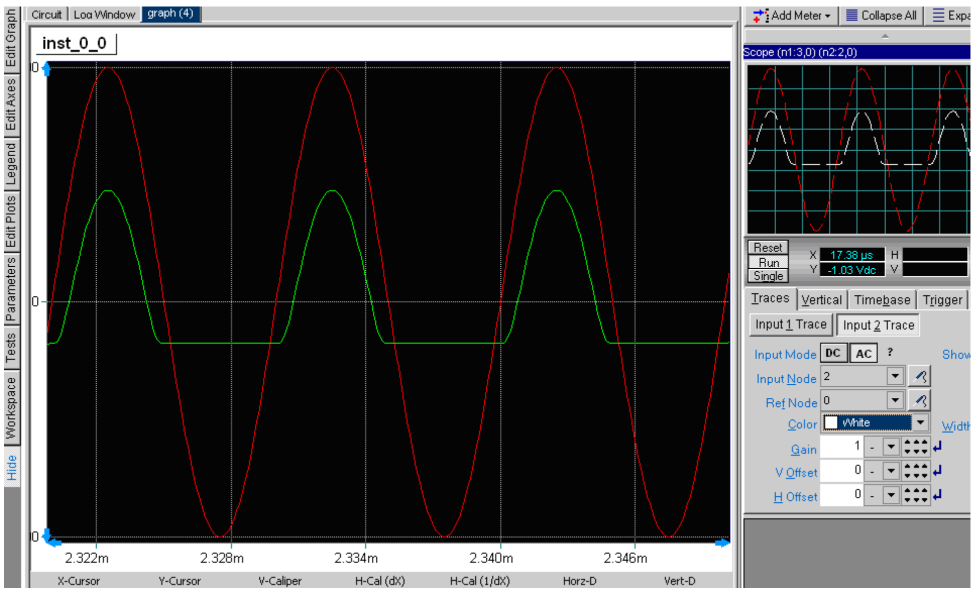


Figure 12.2: Simulation Result for Half Wave Rectifier Circuit

COMMENTS AND CONCLUSIONS

Half-wave rectification is, however, a very simple way to reduce power to a resistive load. Some two-position lamp dimmer switches apply full AC power to the lamp filament for "full" brightness and then half-wave rectify it for a lesser light output. For most power applications, half-wave rectification is insufficient for the task. The harmonic content of the rectifier's output waveform is very large and consequently difficult to filter. Furthermore, the AC power source only supplies power to the load once every half-cycle, meaning that much of its capacity is unused. Fortunately, there is a rectifier in which the AC power source only supplies power to the load for a full cycle utilizing all its capacity. This circuit is known as the full wave rectifier.

EXPERIMENT NO.13

EXPERIMENT NO: 13

NAME

Full Wave Rectifier

OBJECTIVE

To design and simulate a Full Wave Rectifier circuit using B^{2}Spice.

CIRCUIT NOTE

Name	B^{2}Spice Components Used	Description	Number Of Components Required
DIODE	Bav74	D	2
RESISTOR	R	Resistor	1
VIN	Vsin	Ac Voltage Source	2
GND	Gnd	Ground	2

DESIGN AND THEORY

A rectifier is an electrical device that converts alternating current (AC) to direct current (DC), a process known as rectification. Rectifiers have many uses including as components of power supplies and as detectors of radio signals. Rectifiers may be made of solid state diodes, vacuum tube diodes, mercury arc valves, and other components.

When only one diode is used to rectify AC (by blocking the negative or positive portion of the waveform), the difference between the term diode and the term rectifier is merely one of usage, i.e., the term rectifier describes a diode that is being used to convert AC to DC. Almost all rectifiers comprise a number of diodes in a specific arrangement for more efficiently converting AC to DC than is possible with only one diode. Before the development of silicon semiconductor rectifiers, vacuum tube diodes and copper (I) oxide or selenium rectifier stacks were used.

A Full Wave Rectifier is a circuit, which converts an ac voltage into a pulsating dc voltage using both half cycles of the applied ac voltage. It uses two diodes of which one conducts during one half cycle while the other conducts during the other half cycle of the applied ac voltage.

During the positive half cycle of the input voltage, diode D1 becomes forward biased and D2 becomes reverse biased. Hence D1 conducts and D2 remains OFF. The load current flows through D1 and the voltage drop across R_L will be equal to the input voltage.

During the negative half cycle of the input voltage, diode D1 becomes reverse biased and D2 becomes forward biased. Hence D1 remains OFF and D2 conducts. The load current flows through D2 and the voltage drop across R_L will be equal to the input voltage.

The ripple factor for a Full Wave Rectifier is given by

$$\gamma = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1}$$

The average voltage or the dc voltage available across the load resistance is

$$V_{dc} = \frac{1}{\pi}\int_0^{\pi} V_m \sin \omega t \, d(\omega t)$$

$$= \frac{V_m}{\pi}\left[-\cos \omega t\right]_0^{\pi} = \frac{2V_m}{\pi}$$

$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{2V_m}{\pi R_L} = \frac{2I_m}{\pi} \quad and \quad I_{rms} = \frac{I_m}{\sqrt{2}}$$

RMS value of the voltage at the load resistance is

$$V_{rms} = \left[\frac{1}{\pi}\int_0^{\pi} V_m^2 \sin^2 \omega t \, d(\omega t)\right]^{1/2} = \frac{V_m}{\sqrt{2}}$$

$$\therefore \gamma = \sqrt{\left(\frac{V_m/\sqrt{2}}{2V_m/\pi}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{8}\right)^2 - 1} = 0.482$$

Efficiency, η is the ratio of the dc output power to ac input power

$$\eta = \frac{dc\,output\,power}{ac\,input\,power} = \frac{P_{dc}}{P_{ac}}$$

$$\frac{V_{dc}^2/R_L}{V_{rms}^2/R_L} = \frac{\left[2V_m/\pi\right]^2}{\left[V_m/\sqrt{2}\right]^2} = 8/\pi^2 = 0.812 = \underline{\underline{81.2\%}}$$

The maximum efficiency of a Full Wave Rectifier is **81.2%.**

Form factor is defined as the ratio of the rms value of the output voltage to the average value of the output voltage.

$$Form\ factor = \frac{rms\,value\,of\ output\,voltage}{average\,value\,of\ the\ output\,voltage}$$

$$= \frac{\left(V_m/\sqrt{2}\right)}{\left(2V_m/\pi\right)} = \frac{\pi}{2\sqrt{2}} = \underline{\underline{1.11}}$$

Peak factor is defined as the ratio of the peak value of the output voltage to the rms value of the output voltage.

$$Peak\ factor = \frac{peak\,value\,of\ the\ output\,voltage}{rms\,value\,of\ the\ output\,voltage} = \frac{V_m}{\left(V_m/\sqrt{2}\right)} = \underline{\underline{\sqrt{2}}}$$

Peak inverse voltage for Full Wave Rectifier is $2V_m$ because the entire secondary voltage appears across the non-conducting diode.

This concludes the explanation of the various factors associated with Full Wave Rectifier.

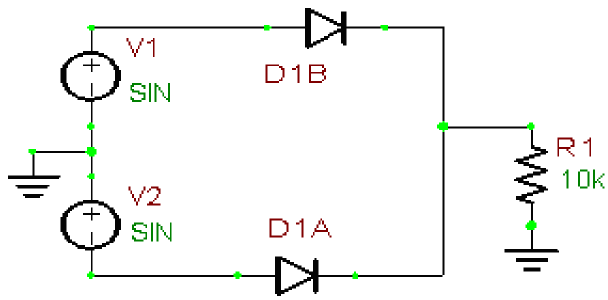


Figure 13.1: Full Wave Rectifier

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 13.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal; for example, by temporarily removing C_S, you can see that the output signal dramatically decreases. By changing the source and drain resistors, it may be possible to further optimize the amplifier for a larger peak output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process. The output waveform may be observed in the Oscilloscope.

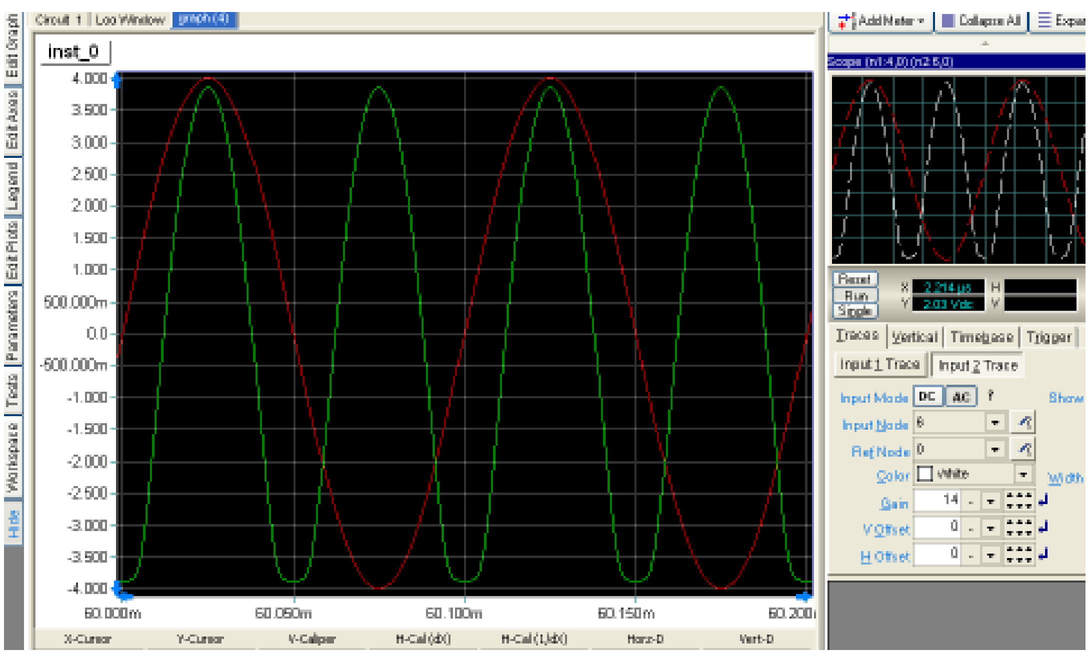


Figure 13.2: Simulation Result for Full Wave Rectifier Circuit

COMMENTS AND CONCLUSIONS

As you may recall from your past studies in electricity, every circuit has advantages and disadvantages. The full-wave rectifier is no exception. In studying the full-wave rectifier, you may have found that by doubling the output frequency, the average voltage has doubled, and the resulting signal is much easier to filter because of the high ripple frequency. The only disadvantage is that the peak voltage in the full-wave rectifier is only half the peak voltage in the half-wave rectifier. Fortunately, there is a rectifier which produces the same peak voltage as a half-wave rectifier and the same ripple frequency as a full-wave rectifier. This circuit is known as the bridge rectifier.

EXPERIMENT NO.14

EXPERIMENT NO:14

NAME

Transistorised Series Voltage Regulators

OBJECTIVE

To design and simulate a Series Voltage Regulator circuit with transistor using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number Of Components Required
RESISTOR	R	Resistor	3
DIODE	Bal74	Diode	1
ZENER DIODE	Bzx55b10	Zener Diode	1
TRANSISTOR	Q2n3904	Transistor	1
VIN	Vsin	Ac Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

The schematic for a typical series voltage regulator is shown in Figure 14.1. Because the total load current passes through this transistor, it is sometimes called a "pass transistor." Other components which make up the circuit are the current limiting resistor (R2) and the Zener diode (X1).

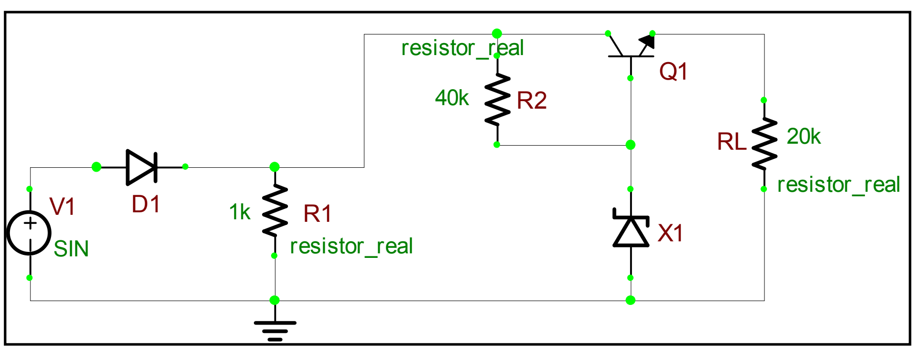


Figure 14.1: Series Voltage Regulator Circuit

Recall that a Zener diode is a diode that block current until a specified voltage is applied. Remember also that the applied voltage is called the breakdown, or Zener voltage. Zener diodes are available with different Zener voltages. When the Zener voltage is reached, the Zener diode conducts from its anode to its cathode (with the direction of the arrow).

In this voltage regulator, Q1 has a constant voltage applied to its base. This voltage is often called the reference voltage. As changes in the circuit output voltage occur, they are sensed at the emitter of Q1 producing a corresponding change in the forward bias of the transistor. In other words, Q1 compensates by increasing or decreasing its resistance in order to change the circuit voltage division.

The Zener used in this regulator is a 5-volt Zener. In this instance the Zener or breakdown voltage is 5 volts. The Zener establishes the value of the base voltage for Q1. The output voltage will equal the Zener voltage minus a 0.7-volt drop across the forward biased base-emitter junction of Q1, or 4.3 volts. Because the output voltage is 4.3 volts, the voltage drop across Q1 must be 15.7 volts. This voltage drop restores the output voltage to 4.3 volts.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 14.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

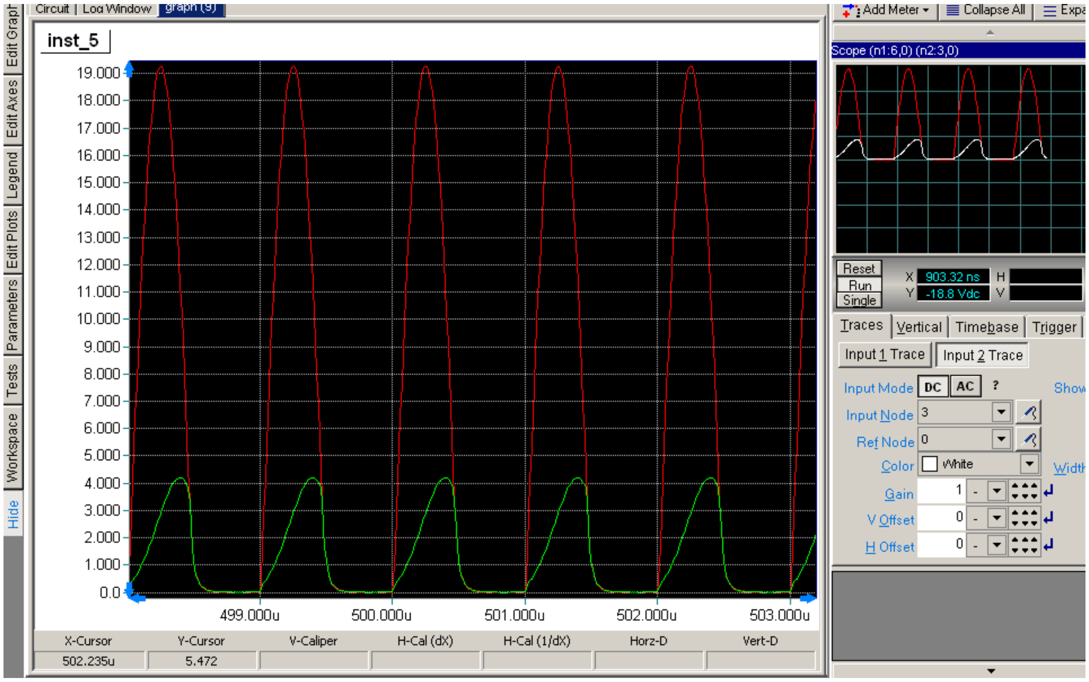


Figure 14.2: Simulation Result for Series Voltage Regulator Circuit

COMMENTS AND CONCLUSION

The entire cycle takes only a fraction of a second and, therefore, the change is not visible on an oscilloscope or readily measurable with other standard test equipment. The output can directly be shown on the oscilloscope. The output that we get is a regulated 4.3 volts signal.

EXPERIMENT NO.15

EXPERIMENT NO:15

NAME

Transistorised Shunt Voltage Regulators

OBJECTIVE

To design and simulate a Shunt Voltage Regulator circuit with transistor using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number Of Components Required
RESISTOR	R	Resistor	3
DIODE	Bal74	Diode	1
ZENER DIODE	Bzx55b10	Zener Diode	1
TRANSISTOR	Q2n3904	Transistor	1
VIN	Vsin	Ac Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

The schematic shown in Figure 1 is that of a shunt voltage regulator. Notice that Q1 is in parallel with the load. Components of this circuit are identical with those of the series voltage regulator except for the addition of fixed resistor R_S. As you study the schematic, you will see that this resistor is connected in series with the output load resistance. The current limiting resistor (R1) and Zener diode (X1) provide a constant reference voltage for the base-collector junction of Q1. Notice that the bias of Q1 is determined by the voltage drop across R_S and R1. As you should know, the amount of forward bias across a transistor affects its total resistance. In this case, the voltage drop across R_S is the key to the total circuit operation.

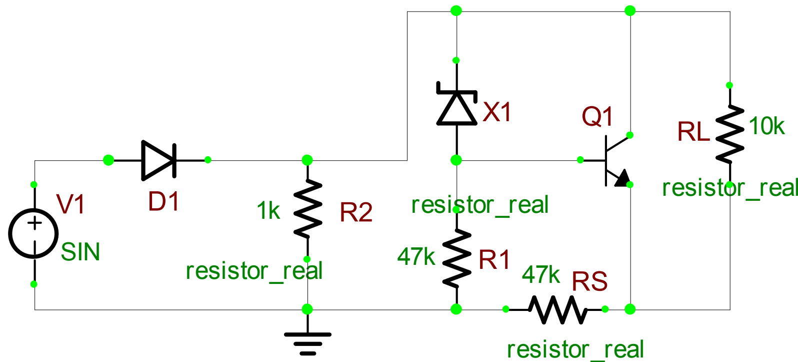


Figure 15.1: Shunt Voltage Regulator Circuit

Figure 15.1 is the schematic for a typical shunt-type regulator. The voltage drop across the Zener diode (X1) remains constant at 5.6 volts. This means that with a 20-volt input voltage, the voltage drop across R1 is 14.4 volts. With a base-emitter voltage of 0.7 volt, the output voltage is equal to the sum of the voltages across X1 and the voltage at the base-emitter junction of Q1. In this example, with an output voltage of 6.1 volts and a 20-volt input voltage, the voltage drop across R_S equals 13.9 volts. Study the schematic to understand fully how these voltages are developed.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 15.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

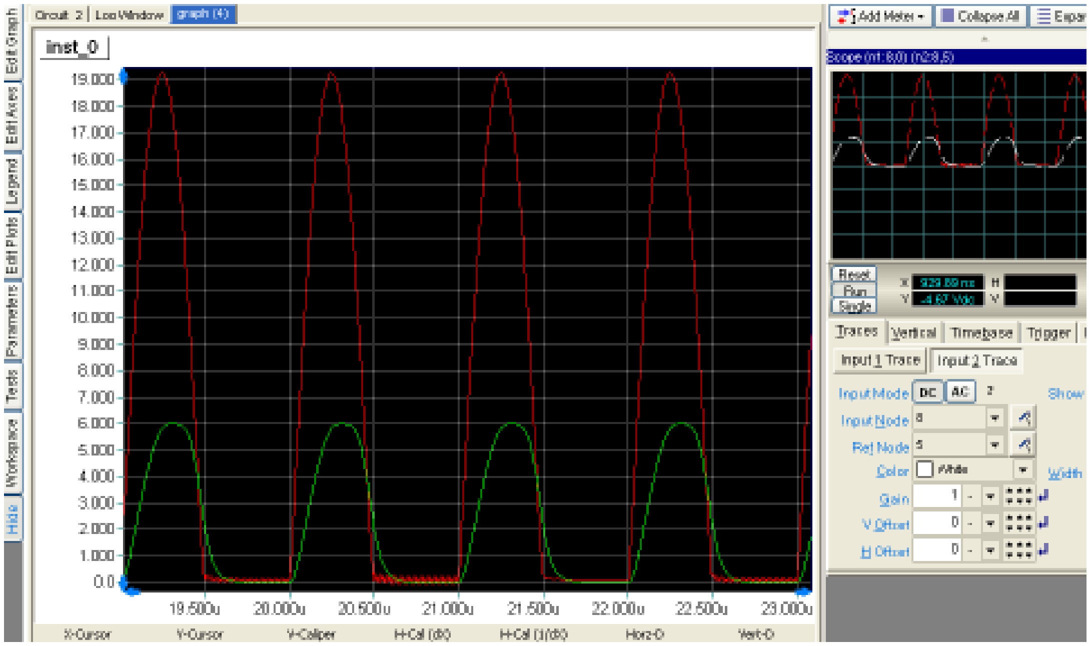


Figure 15.2: Simulation Result for Shunt Voltage Regulator Circuit

COMMENTS AND CONCLUSION

The entire cycle takes only a fraction of a second and, therefore, the change is not visible on an oscilloscope or readily measurable with other standard test equipment. The output can directly be shown on the oscilloscope. The output that we get is a regulated 6.1 volts signal.

EXPERIMENT NO.16

EXPERIMENT NO: 16

NAME

Digital Counter

OBJECTIVE

The purpose of this experiment is to investigate the operation of an IC based counter circuit using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number Of Components Required
COUNTER IC	74393	Counter	1
SWITCH	Toggle	Toggle Switch	1
CLOCK	Clock Digital	Clock Source	1

DESIGN AND THEORY

One common requirement in digital circuits is counting, both forward and backward. Digital clocks and watches are everywhere, timers are found in a range of appliances from microwave ovens to VCRs, and counters for other reasons are found in everything from automobiles to test equipment.

Although we will see many variations on the basic counter, they are all fundamentally very similar. The demonstration below shows the most basic kind of binary counting circuit.

In the 4-bit counter to the right, we are using edge-triggered master-slave flip-flops similar to those in the Sequential portion of these pages. The output of each flip-flop changes state on the falling edge (1-to-0 transistion) of the T input.

The count held by this counter is read in the reverse order from the order in which the flip-flops are triggered. Thus, output D is the high order of the count, while output A is the low order. The binary count held by the counter is then DCBA, and runs from 0000 (decimal 0) to 1111 (decimal 15). The next clock pulse will cause the counter to try to increment to 10000 (decimal 16). However, that 1 bit is not held by any flip-flop and is therefore lost. As a result, the counter actually reverts to 0000, and the count begins again.

A major problem with the counters shown on this page is that the individual flip-flops do not all change state at the same time. Rather, each flip-flop is used to trigger the next one in the series. Thus, in switching from all 1s (count = 15) to all 0s (count wraps back to 0), we don't see a smooth transition. Instead, output A falls first, changing the apparent count to 14. This triggers output B to fall, changing the apparent count to 12. This in turn triggers output C, which leaves a count of 8 while triggering output D to fall. This last action finally leaves us with the correct output count of zero. We say that the change of state "ripples" through the counter from one flip-flop to the next. Therefore, this circuit is known as a "ripple counter."

This causes no problem if the output is only to be read by human eyes; the ripple effect is too fast for us to see it. However, if the count is to be used as a selector by other digital circuits (such as a multiplexer or demultiplexer), the ripple effect can easily allow signals to get mixed together in an undesirable fashion. To prevent this, we need to devise a method of causing all of the flip-flops to change state at the same moment. That would be known as a "synchronous counter" because the flip-flops would be synchronized to operate in unison. That is the subject of the next page in this series.

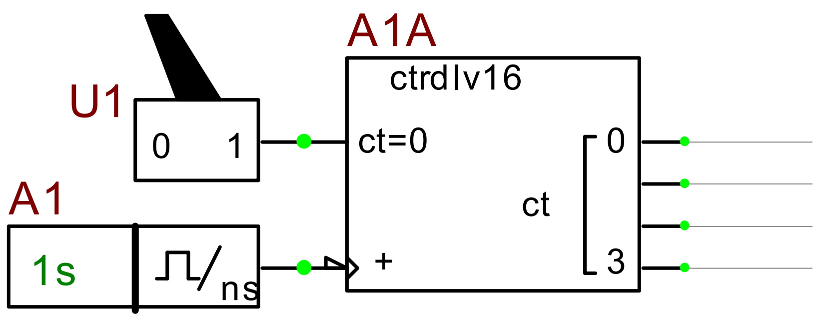


Figure 16.1: IC Based Counter

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 16.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit. Here the step 7 and Step 8 can be omitted and then in Step 9 directly click on the Run button in order to see the counter running.

Connect the amplifier’s input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit’s component values, and see how this affects the output signal; for example, by changing the clock period, you can see that the output dramatically changes. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

COMMENTS AND CONCLUSIONS

The Counter above is driven by the digital clock available in B^2Spice V5. Thus in this way counter can be simulated using B^2Spice V5.

EXPERIMENT NO.17

EXPERIMENT NO: 17

NAME

Multiplexer

OBJECTIVE

The purpose of this experiment is to investigate the operation of an IC based multiplexer circuit using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number Of Components Required
MULTIPLEXER IC	74353_Ls	Multiplexer	1
SWITCH	Toggle	Toggle Switch	1
INPUT	Input_Port	Input Port	6
OUTPUT	Output_Port	Output Port	1

DESIGN AND THEORY

A multiplexer is a combinatorial circuit that is given a certain number (usually a power of two) data inputs, let us say 2^n, and n address inputs used as a binary number to select one of the data inputs. The multiplexer has a single output, which has the same value as the selected data input.

In other words, the multiplexer works like the input selector of a home music system. Only one input is selected at a time, and the selected input is transmitted to the single output. While on the music system, the selection of the input is made manually, the multiplexer chooses its input based on a binary number, the address input.

The truth table for a multiplexer is huge for all but the smallest values of n. We therefore use an abbreviated version of the truth table in which some inputs are replaced by `-' to indicate that the input value does not matter. Here is such a truth table for n = 2.

A	B	D_3	D_2	D_1	D_0	|	F
-	-	-	-	-	-	---	-
0	0	-	-	-	0	|	0
0	0	-	-	-	1	|	1
0	1	-	-	0	-	|	0
0	1	-	-	1	-	|	1

1	0	-	0	-	-	|	0
1	0	-	1	-	-	|	1
1	1	0	-	-	-	|	0
1	1	1	-	-	-	|	1

The same way we can simplify the truth table for the multiplexer, we can also simplify the corresponding circuit. Indeed, our simple design method would yield a very large circuit. The simplified IC based circuit and equivalent logic circuit looks like this:

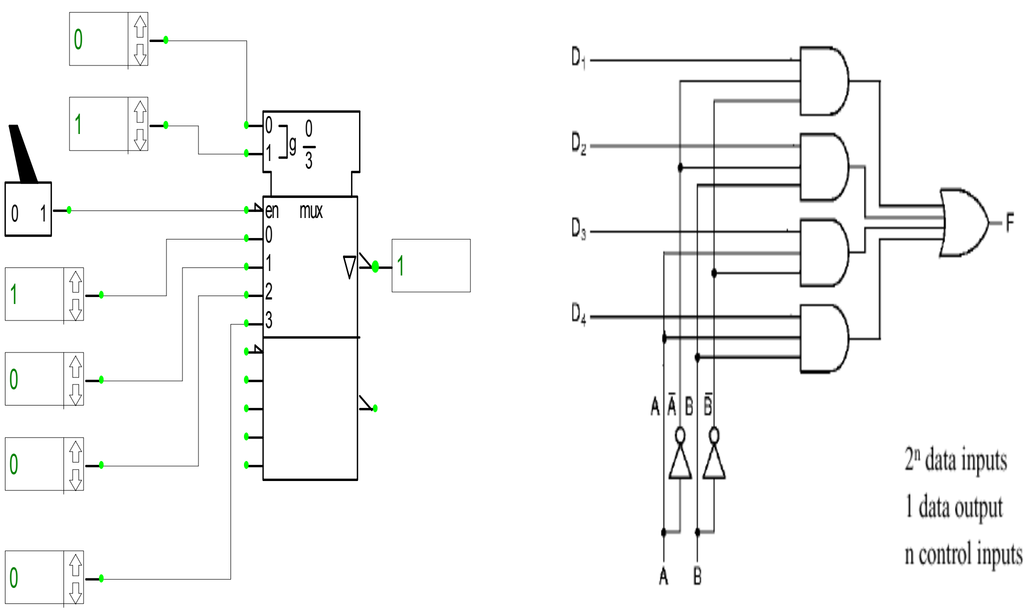


Figure 17.1: a) IC Based 4:1 Multiplexer. b) 4:1 Multiplexer circuit.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 17.1 (a) using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit. Here the step 7 and Step 8 can be omitted and then in Step 9 directly click on the Run button in order to verify for the multiplexer functionality.

Connect the amplifier’s input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit’s component values, and see how this affects the output signal; for example, by changing the input values and the select lines at the input_ports, you can see that the output immediately changes. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

COMMENTS AND CONCLUSIONS

The Counter above is driven by the digital clock available in B^{2}Spice V5. Thus in this way counter can be simulated using B^{2}Spice V5.

EXPERIMENT NO.18

EXPERIMENT NO: 18

NAME

555 Timer

OBJECTIVE

The purpose of this experiment is to investigate the operation of an IC based 555 Timer circuit using B2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number Of Components Required
555 IC	Tlc555d	555 Timer Ic	1
RESISTOR	Res	Resistor	2
CAPACITOR	Cap	Capacitor	2
VDC	Vcc	Dc Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

The 555 timer IC is an amazingly simple yet versatile device. It has been around now for many years and has been reworked into a number of different technologies. The two primary versions today are the original bipolar design and the more recent CMOS equivalent. These differences primarily affect the amount of power they require and their maximum frequency of operation; they are pin-compatible and functionally interchangeable.

This page contains only a description of the 555 timer IC itself. Functional circuits and a few of the very wide range of its possible applications will be covered in additional pages in this category.

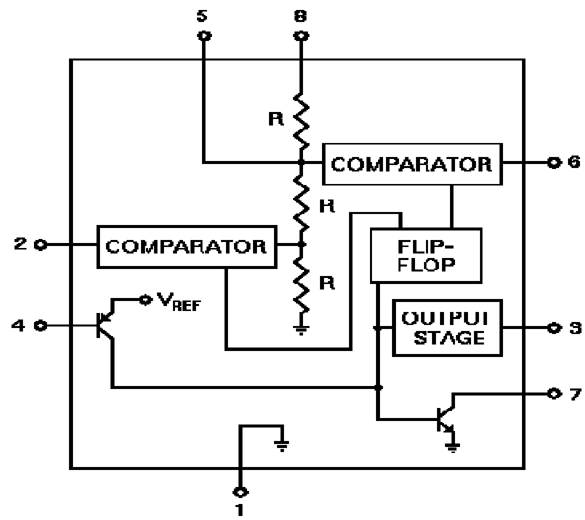


Figure 18.1: Block Diagram of 555 Timer

The figure to the right shows the functional block diagram of the 555 timer IC. The IC is available in either an 8-pin round TO3-style can or an 8-pin mini-DIP package. In either case, the pin connections are as follows:

- Ground.
- Trigger input.
- Output.
- Reset input.
- Control voltage.
- Threshold input.
- Discharge.
- $+V_{CC}$. +5 to +15 volts in normal use.

The operation of the 555 timer revolves around the three resistors that form a voltage divider across the power supply, and the two comparators connected to this voltage divider. The IC is quiescent so long as the trigger input (pin 2) remains at $+V_{CC}$ and the threshold input (pin 6) is at ground. Assume the reset input (pin 4) is also at $+V_{CC}$ and therefore inactive, and that the control voltage input (pin 5) is unconnected. Under these conditions, the output (pin 3) is at ground and the discharge transistor (pin 7) is turned on, thus grounding whatever is connected to this pin.

The three resistors in the voltage divider all have the same value (5K in the bipolar version of this IC), so the comparator reference voltages are 1/3 and 2/3 of the supply voltage, whatever that may be. The control voltage input at pin 5 can directly affect this relationship, although most of the time this pin is unused.

The internal flip-flop changes state when the trigger input at pin 2 is pulled down below $+V_{CC}/3$. When this occurs, the output (pin 3) changes state to $+V_{CC}$ and the discharge transistor (pin 7) is turned off. The trigger input can now return to $+V_{CC}$; it will not affect the state of the IC.

However, if the threshold input (pin 6) is now raised above $(2/3)+V_{CC}$, the output will return to ground and the discharge transistor will be turned on again. When the threshold input returns to ground, the IC will remain in this state, which was the original state when we started this analysis.

The easiest way to allow the threshold voltage (pin 6) to gradually rise to (2/3)+V_{CC} is to connect it to a capacitor being allowed to charge through a resistor. In this way we can adjust the R and C values for almost any time interval we might want.

The 555 can operate in either monostable or astable mode, depending on the connections to and the arrangement of the external components. Thus, it can either produce a single pulse when triggered, or it can produce a continuous pulse train as long as it remains powered.

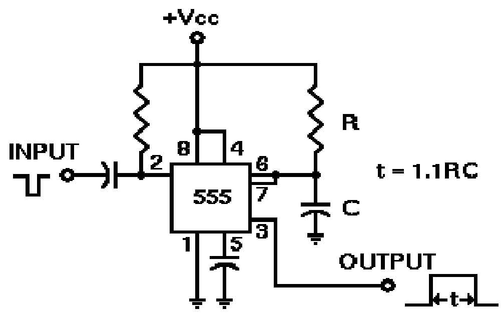


Figure 18.2: Monostable Mode

In monostable mode, the timing interval, t, is set by a single resistor and capacitor, as shown to the right. Both the threshold input and the discharge transistor (pins 6 & 7) are connected directly to the capacitor, while the trigger input is held at +V_{CC} through a resistor. In the absence of any input, the output at pin 3 remains low and the discharge transistor prevents capacitor C from charging.

When an input pulse arrives, it is capacitively coupled to pin 2, the trigger input. The pulse can be either polarity; its falling edge will trigger the 555. At this point, the output rises to +V_{CC} and the discharge transistor turns off. Capacitor C charges through R towards +V_{CC}. During this interval, additional pulses received at pin 2 will have no effect on circuit operation.

The standard equation for a charging capacitor applies here: $e = E\,(1 - \varepsilon^{(-t/RC)})$. Here, "e" is the capacitor voltage at some instant in time, "E" is the supply voltage, V_{CC}, and "ε" is the base for natural logarithms, approximately 2.718. The value "t" denotes the time that has passed, in seconds, since the capacitor started charging.

We already know that the capacitor will charge until its voltage reaches (2/3)+V_{CC}, whatever that voltage may be. This doesn't give us absolute values for "e" or "E," but it does give us the ratio e/E = 2/3. We can use this to compute the time, t, required to charge capacitor C to the voltage that will activate the threshold comparator:

$$2/3 = 1 - \varepsilon^{(-t/RC)}$$
$$-1/3 = -\varepsilon^{(-t/RC)}$$
$$1/3 = \varepsilon^{(-t/RC)}$$
$$\ln(1/3) = -t/RC$$
$$-1.0986123 = -t/RC$$
$$t = 1.0986123RC$$
$$t = 1.1RC$$

The value of 1.1RC isn't exactly precise, of course, but the round off error amounts to about 0.126%, which is much closer than component tolerances in practical circuits, and is very easy to use. The values of R and C must be given in Ohms and Farads, respectively, and the time will be in seconds. You can scale the values as needed and appropriate for your application, provided you keep proper track of your powers of 10. For example, if you specify R in mega ohms and C in microfarads, t will still be in seconds. But if you specify R in kilo ohms and C in microfarads, t will be in milliseconds. It's not difficult to keep track of this, but you must be sure to do it accurately in order to correctly calculate the component values you need for any given time interval.

The timing interval is completed when the capacitor voltage reaches the (2/3)+V_{CC} upper threshold as monitored at pin 6. When this threshold voltage is reached, the output at pin 3 goes low again, the discharge capacitor (pin 7) is turned on, and the capacitor rapidly discharges back to ground once more. The circuit is now ready to be triggered once again.

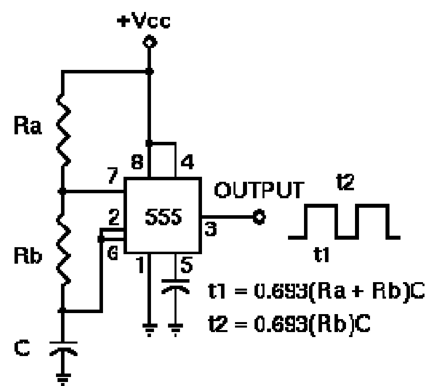


Figure 18.3: Astable Mode

If we rearrange the circuit slightly so that both the trigger and threshold inputs are controlled by the capacitor voltage, we can cause the 555 to trigger itself repeatedly. In this case, we need two resistors in the capacitor charging path so that one of them can also be in the capacitor discharge path. This gives us the circuit shown to the left.

In this mode, the initial pulse when power is first applied is a bit longer than the others, having a duration of 1.1(Ra + Rb)C. However, from then on, the capacitor alternately charges and discharges between the two comparator threshold voltages. When charging, C starts at (1/3)V_{CC} and charges towards V_{CC}. However, it is interrupted exactly halfway there, at (2/3)V_{CC}.

Therefore, the charging time,

t1 = ln(1/2)(Ra + Rb)C = 0.693(Ra + Rb)C.

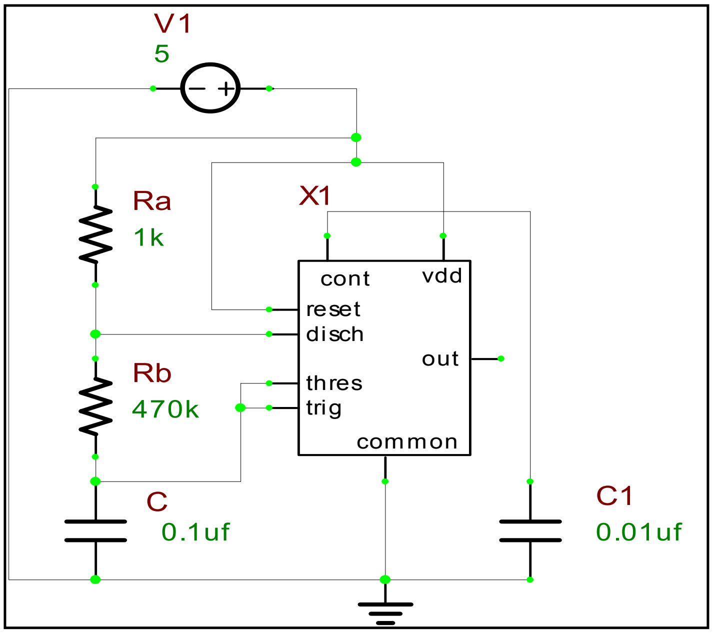
When the capacitor voltage reaches (2/3)V_{CC}, the discharge transistor is enabled (pin 7), and this point in the circuit becomes grounded. Capacitor C now discharges through Rb alone. Starting at (2/3)V_{CC}, it discharges towards ground, but again is interrupted halfway there, at (1/3)V_{CC}. The discharge time, t2, then, is -ln(1/2)(Rb)C = 0.693(Rb)C.

The total period of the pulse train is t1 + t2, or 0.693(Ra + 2Rb)C. The output frequency of this circuit is the inverse of the period, or 1.44/(Ra + 2Rb)C.

Note that the duty cycle of the 555 timer circuit in astable mode cannot reach 50%. On time must always be longer than off time, because Ra must have a resistance value greater than zero to prevent the discharge transistor from directly shorting V_{CC} to ground. Such an action would immediately destroy the 555 IC.

One interesting and very useful feature of the 555 timer in either mode is that the timing interval for either charge or discharge is independent of the supply voltage, V_{CC}. This is because the same V_{CC} is used both as the charging voltage and as the basis of the reference voltages for the two comparators inside the 555. Thus, the timing equations above depend only on the values for R and C in either operating mode.

In addition, since all three of the internal resistors used to make up the reference voltage divider are manufactured next to each other on the same chip at the same time, they are as nearly identical as can be. Therefore, changes in temperature will also have very little effect on the timing intervals, provided the external components are temperature stable. A typical commercial 555 timer will show a drift of 50 parts per million per Centigrade degree of temperature change (50 ppm/°C) and 0.01%/Volt



change in V_{CC}. This is negligible in most practical applications.

Figure 18.4: IC Based 555 Timer Circuit

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 18.4 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

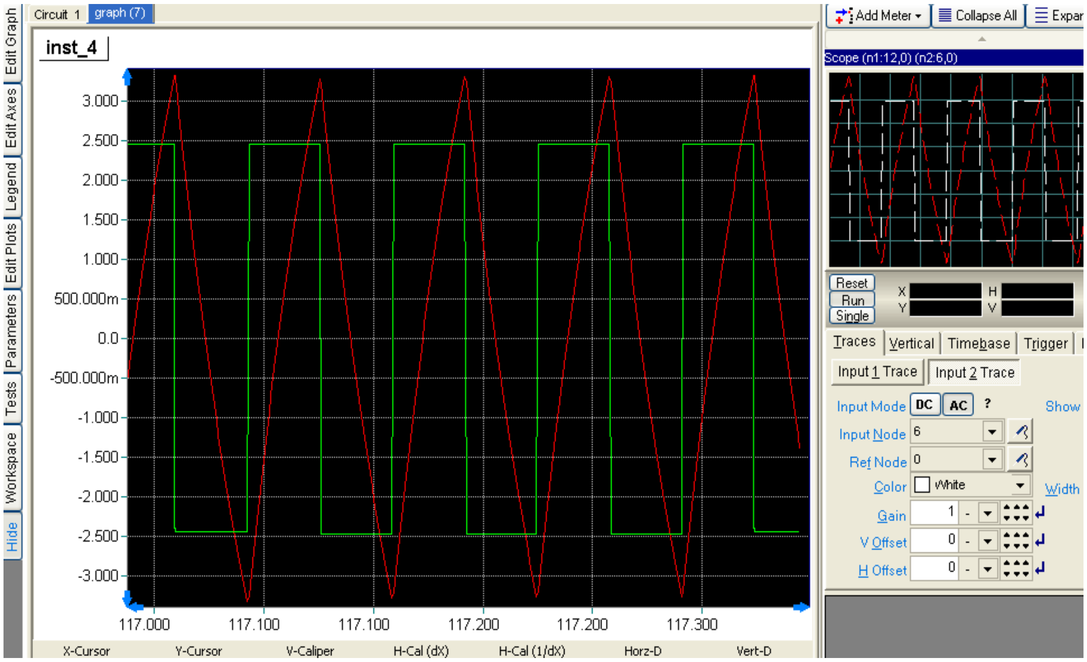


Figure 18.5: Simulation Result for IC Based 555 Timer Circuit

COMMENTS AND CONCLUSIONS

The Counter above is driven by the digital clock available in B^2Spice V5. Thus the counter can be designed and simulated using B^2Spice V5.

EXPERIMENT
NO.19

EXPERIMENT NO: 19

NAME

LC Oscillator

OBJECTIVE

To Design and Simulate a LC Oscillator circuit using B^2Spice.

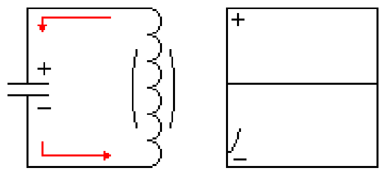
CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number Of Components Required
BJT	Q2n3904	Transistor	1
RESISTOR	Res	Resistor	5
CAPACITOR	Cap	Capacitor	5
VDC	Vcc	Dc Voltage Source	1
GND	Gnd	Ground	1

DESIGN AND THEORY

Electronic oscillators are commonly found in everyday circuits, ranging from antique radios to the transmitter in your TV remote. The basic job of an electronic oscillator is to generate an oscillating output often at a constant frequency. Various outputs can be sine, square, saw-tooth, triangle, or complex waveforms. Here a basic sinusoidal LC oscillator is explained.

This oscillator consists of a capacitor and a coil connected in parallel. To understand how the LC oscillator basically works, let's start off with the basics. Suppose a capacitor is charged by a battery. Once the capacitor is charged, one plate of the capacitor has more electrons than the other plate, thus it is charged. Now, when it is discharged through a wire, the electrons return to the positive plate, thus making the capacitor's plates neutral, or discharged. However, this action works differently when you discharge a capacitor through a coil. When current is applied through a coil, a magnetic field is generated around the coil. This magnetic field generates a voltage across the coil that opposes the direction of electron flow. Because of this, the capacitor does not discharge right away. The smaller the coil, the faster the capacitor discharges. Now the interesting part happens. Once the capacitor is fully discharged through the coil, the magnetic field starts to collapse around the coil. The voltage induced from the collapsing magnetic field recharges the capacitor oppositely. Then the capacitor begins to discharge through the coil again, generating a magnetic field. This process continues until the capacitor is completely discharged due to resistance.



Technically this basic LC circuit generates a sine wave that loses voltage in every cycle. To overcome this, additional voltage is applied to keep the oscillator from losing voltage. However, to keep this oscillator going well, a switching method is used. A vacuum tube (or a solid-state equivalent such as a FET) is used to keep this LC circuit oscillating. The advantage of using a vacuum tube is that they can oscillate at specified frequencies such as a thousand cycles per second.

There are several different types of LC oscillators. A little off subject; one well known and entertaining oscillator is known as the tesla coil, which uses an unique LC circuit using the spark gap to oscillate.

The Colpitts oscillator is a type of LC oscillator, which has tapped capacitance.

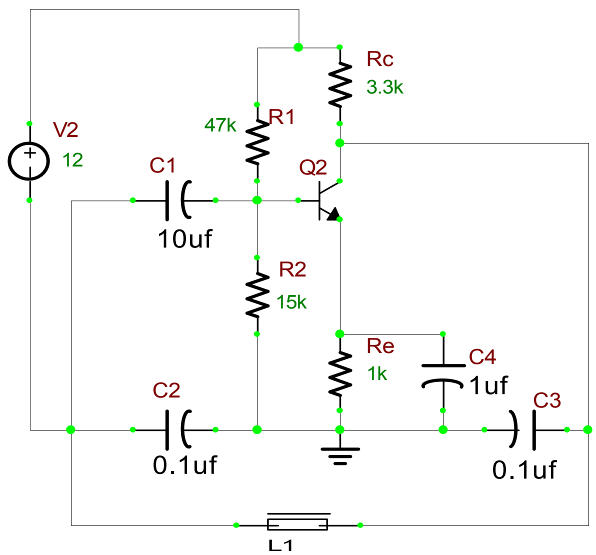


Figure 19.1: LC Oscillator Circuit (Colpitt's Oscillator)

The tap between the two capacitors is grounded and the feedback is obtained from the coupling capacitor, C1. The amount of feedback depends on the ratio of C2 to C3. The capacitor part of the LC circuit consists of both C2 and C3, which determines the oscillating frequency. This oscillator has more frequency stabilities than other types of LC oscillators such as Hartley oscillator.

COMPUTER SIMULATION USING B^{2}SPICE V5

Draw the circuit shown in Figure 19.1 using the editor within B^{2}Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B^{2}Spice V5. Then connect the oscilloscope in B^{2}Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B^{2}Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process.

The simulation result here is a continuous damping process where we can show them in 2 or 3 consecutive snaps of the simulation process. The below are the result of an oscillation due to the RC Phase Shift Oscillator.

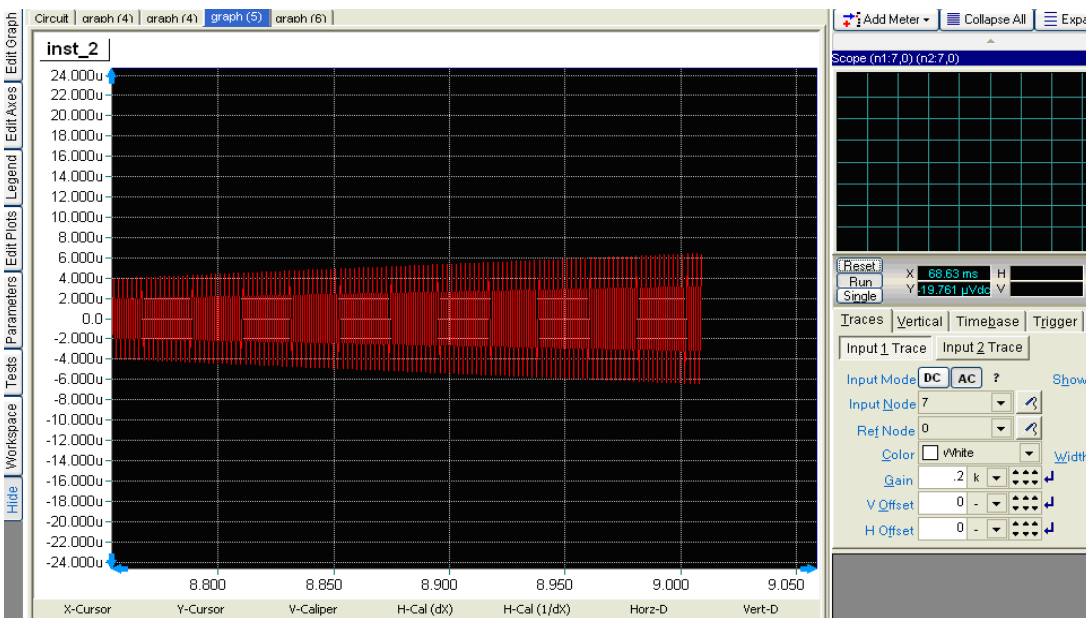


Figure 19.2: a. Simulation Result for LC Oscillator Circuit

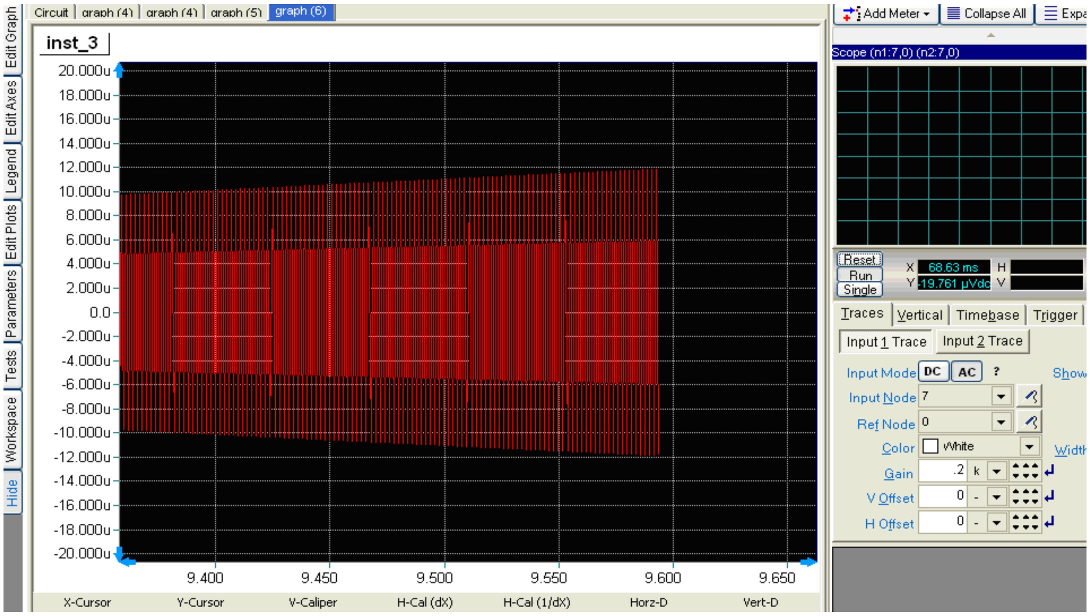


Figure 19.2: b. Simulation Result for LC Oscillator Circuit

COMMENTS AND CONCLUSIONS

Now you have learned about LC oscillator, you can build your own for several purposes such as a radio, a tube tesla coil, a low-power transmitter, or more. There are many solid-state equivalents to the oscillators explained here in which the tube is substituted with a transistor or a FET. Nevertheless, the circuitry is still used in the same manner as their solid-state equivalents.

EXPERIMENT NO.20

EXPERIMENT NO: 20

NAME

Three Phase Full Converter

OBJECTIVE

To design and simulate a Fully Controlled full Bridge Converter circuit using B^2Spice.

CIRCUIT NOTE

Name	B^2Spice Components Used	Description	Number Of Components Required
SCR	Generic	Thyristor	6
RESISTOR	R	Resistor	1
INDUCTOR	L	Inductor	1
VIN	Vsin	Ac Voltage Source	3
GND	Gnd	Ground	1

DESIGN AND THEORY

A 3 Phase Fully Controlled Full Wave Bridge Converter is known as a 6-pulse converter.

It usually is used in industrial applications up to 120kW output power. Two quadrant operation is possible with such converters.

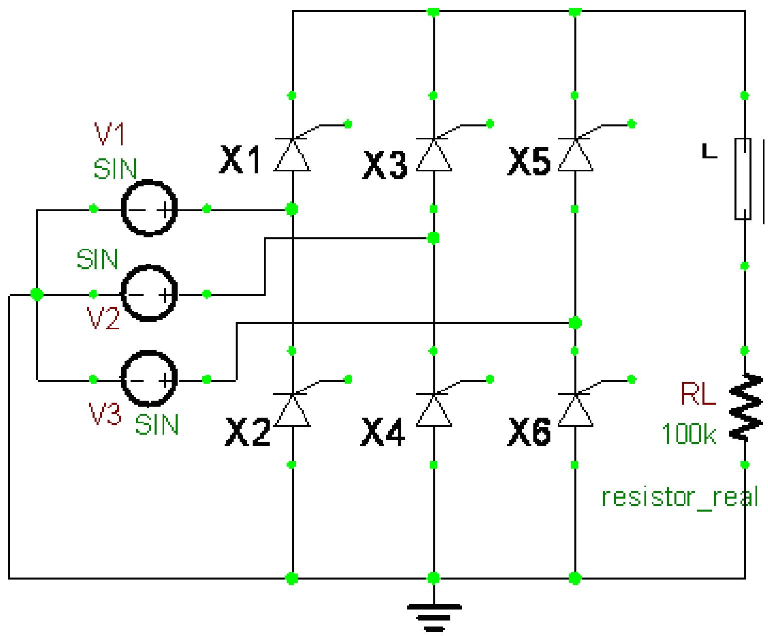
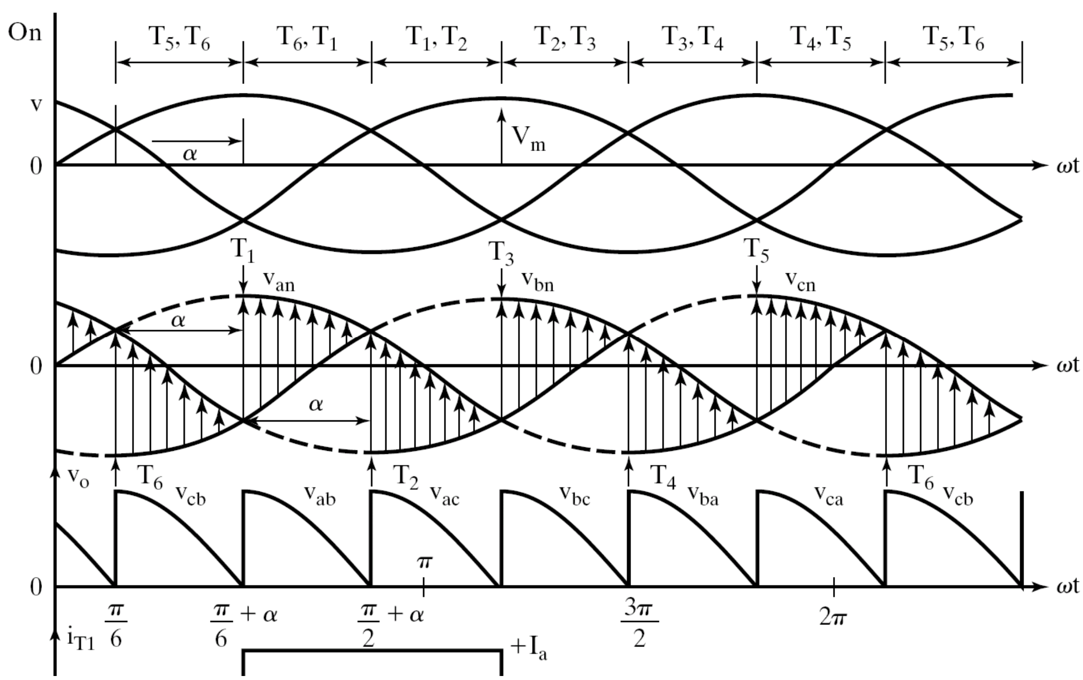


Figure 20.1: 3Φ Fully Controlled Full Bridge Converter



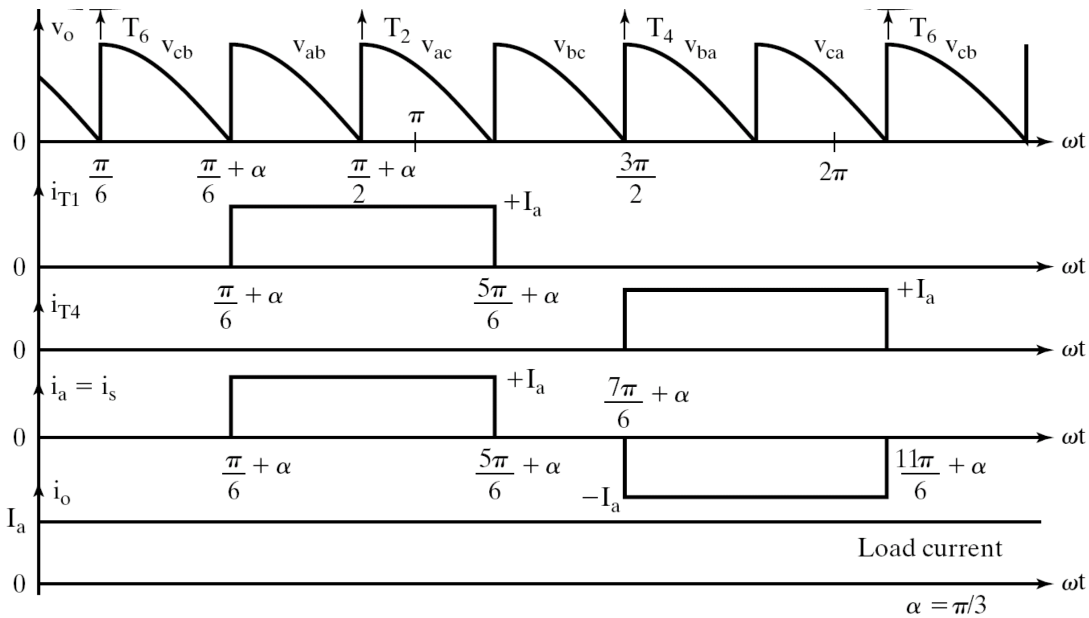


Figure 20.2: Characteristics of a 3Φ Full Converter

The thyristors are triggered at an interval of π / 3. The frequency of output ripple voltage is 6fS. T1 is triggered at ωt = (π/6 + α), T6 is already conducting when T1 is turned ON. During the interval (π/6 + α) to (π/2 + α), T1 and T6 conduct together & the output load voltage is equal to vab = (van – vbn)

T2 is triggered at ωt = (π/2 + α), T6 turns off naturally as it is reverse biased as soon as T2 is triggered. During the interval (π/2 + α) to (5π/6 + α), T1 and T2 conduct together & the output load voltage

vO = vac = (van – vcn). Thyristors are numbered in the order in which they are triggered.

The thyristor triggering sequence is:

12, 23, 34, 45, 56, 61, 12, 23, 34, and so on.

We define three Line Neutral Voltages (3 phase voltages) as follows,

$$v_{RN} = v_{an} = V_m \sin \omega t \quad ; \qquad V_m = \text{ Max. Phase Voltage}$$

$$v_{YN} = v_{bn} = V_m \sin\left(\omega t - \frac{2\pi}{3}\right) = V_m \sin\left(\omega t - 120^0\right)$$

$$v_{BN} = v_{cn} = V_m \sin\left(\omega t + \frac{2\pi}{3}\right) = V_m \sin\left(\omega t + 120^0\right)$$

$$= V_m \sin\left(\omega t - 240^0\right)$$

V_m is the peak phase voltage of a wye-connected source.

The corresponding line-to line Voltages are:

$$v_{RY} = v_{ab} = \left(v_{an} - v_{bn}\right) = \sqrt{3}V_m \sin\left(\omega t + \frac{\pi}{6}\right)$$

$$v_{YB} = v_{bc} = \left(v_{bn} - v_{cn}\right) = \sqrt{3}V_m \sin\left(\omega t - \frac{\pi}{2}\right)$$

$$v_{BR} = v_{ca} = \left(v_{cn} - v_{an}\right) = \sqrt{3}V_m \sin\left(\omega t + \frac{\pi}{2}\right)$$

To derive an expression for the average output voltage of 3-phase full converter with highly inductive load assuming continuous and constant load current the steps are discussed below.

The output load voltage consists of 6 voltage pulses over a period of 2π radians, Hence the average output voltage is calculated as

$$V_{O(dc)} = V_{dc} = \frac{6}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} v_O . d\omega t \quad ;$$

$$v_O = v_{ab} = \sqrt{3}V_m \sin\left(\omega t + \frac{\pi}{6}\right)$$

$$V_{dc} = \frac{3}{\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} \sqrt{3}V_m \sin\left(\omega t + \frac{\pi}{6}\right) . d\omega t$$

$$V_{dc} = \frac{3\sqrt{3}V_m}{\pi}\cos\alpha = \frac{3V_{mL}}{\pi}\cos\alpha$$

Where $V_{mL} = \sqrt{3}V_m =$ Max. line-to-line supply voltage

The Maximum average DC output voltage is obtained for a delay angle,

$\alpha = 0$,

$$V_{dc(\max)} = V_{dm} = \frac{3\sqrt{3}V_m}{\pi} = \frac{3V_{mL}}{\pi}$$

The normalized average dc output voltage is

$$V_{dcn} = V_n = \frac{V_{dc}}{V_{dm}} = \cos\alpha$$

The rms value of the output voltage is found from

$$V_{O(rms)} = \left[\frac{6}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} v_O^2 . d\left(\omega t\right)\right]^{\frac{1}{2}}$$

$$V_{O(rms)} = \left[\frac{6}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} v_{ab}^2 . d\left(\omega t\right) \right]^{\frac{1}{2}}$$

$$V_{O(rms)} = \left[\frac{3}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} 3V_m^2 \sin^2\left(\omega t + \frac{\pi}{6}\right) . d\left(\omega t\right) \right]^{\frac{1}{2}}$$

$$V_{O(rms)} = \sqrt{3} V_m \left(\frac{1}{2} + \frac{3\sqrt{3}}{4\pi} \cos 2\alpha \right)^{\frac{1}{2}}$$

COMPUTER SIMULATION USING B²SPICE V5

Draw the circuit shown in Figure 20.1 using the editor within B²Spice V5. Refer to the steps mentioned in Experiment 1 to design the circuit.

Connect the amplifier's input to the AC voltage source in B²Spice V5. Then connect the oscilloscope in B²Spice V5 to various test points in the circuit, and compare these "measured" waveforms to your predicted and experimentally measured values.

Finally, try changing some of the circuit's component values, and see how this affects the output signal. In summary, by using B²Spice V5, note how easy it is to interactively change any component at will, and immediately see the effects, which further stimulates, encourages, and enhances the learning process. The output waveform may be observed in the Oscilloscope.

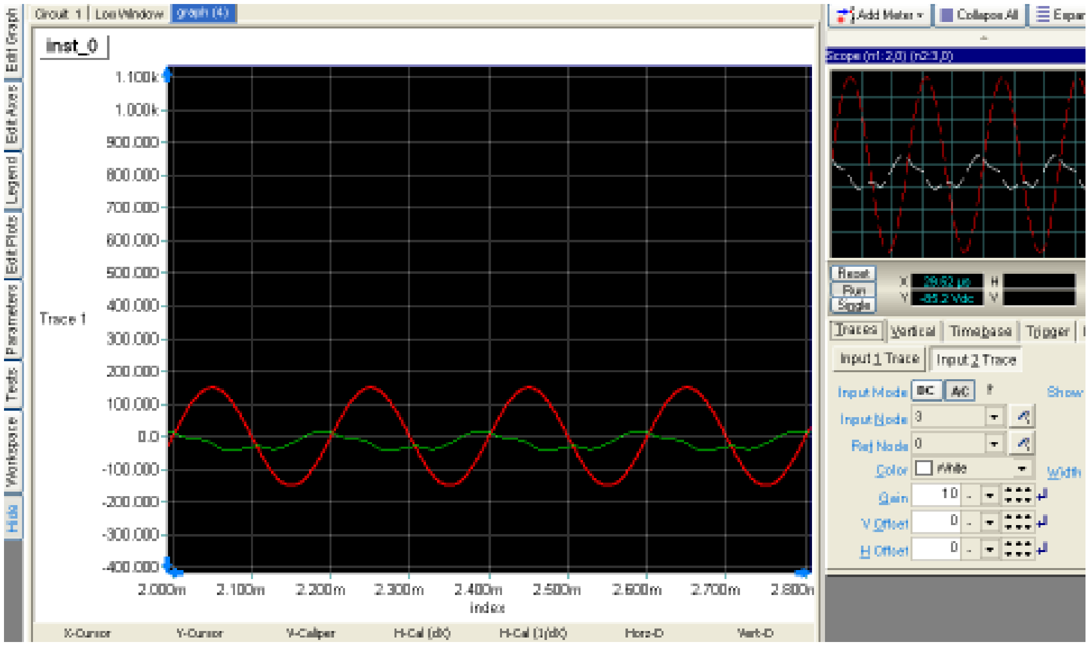


Figure 20.1: 3Φ Fully Controlled Full Bridge Converter

COMMENTS AND CONCLUSIONS

Hence the performance of a 3Φ Fully Controlled Full Bridge Converter can be investigated from the above simulation result. The changes in the configuration to make it a 3Φ Half Controlled Full Bridge Converter and 3Φ Dual Converter can also be done and compared using B^{2}Spice V5.